# HPM5300 Series
# High Performance Microcontroller Datasheet

Rev0.12

- 32-bit RISC-V Processor
  - Supports RV32-IMAFDCBP instruction set
  - 3.57 Coremark/MHz, 1.98 DMIPS/MHz
  - DSP unit with support for SIMD and DSP instructions
  - 16KB L1 instruction cache and 16KB L1 data cache
  - 128KB Instruction Local Memory (ILM) and 128KB Data Local Memory (DLM)
- On-chip Memory
  - Total 288 KB on-chip SRAM, including general-purpose memory and CPU local memories
  - 1 MB built-in flash memory
  - 4096-bit OTP memory
  - 128 KB BOOT ROM
- Power and Clock
  - Multiple on-chip power supplies, including DCDC and LDO
  - Low-power modes: Run mode, Wait mode, Stop mode, Sleep mode and Power-off mode
  - 24MHz crystal oscillator
  - 24MHz and 32KHz internal RC oscillators
  - 2 PLLs, supporting fractional frequency division and spread spectrum
- External Memory Interface
  - 1 XPI serial bus controller, supports various external serial Flash and PSRAM devices
- Motion Control System
  - 2 PWM timers
  - 2 Quadrature Encoder Input (QEIv2) and 2 Quadrature Encoder Output (QEO) interfaces
  - 2 Serial Encoder Interfaces (SEI)
  - 2 Motion Management Controllers (MMC)
  - 1 Resolver-to-Digital Converter (RDC)
  - 1 Programmable Logic Block (PLB)
- Timers
  - 5 groups of 32-bit general-purpose timers
  - 3 watchdog timers
  - Real-Time Clock (RTC)
- Communication Interfaces
  - 9 UARTs, 4 SPIs, 4 I2Cs
  - 1 USB 2.0 OTG with integrated HS-PHY
  - 4 CAN controllers supporting CAN-FD
- High-performance Analog Peripherals
  - 2 ADCs, 16-bit resolution @2MSPS, configurable to 12-bit resolution @4MSPS, supporting up to 16 analog input pins

  - 2 12-bit DACs @1MSPS
  - 2 analog comparators
  - 2 operational amplifiers
- Input/Output
  - 56 GPIOs
  - I/Os support 3.3V and 1.8V operation
- Information Security
  - AES-128/256 encryption/decryption engine, supporting ECB and CBC modes
  - Supports SM2, SM3, SM4 cryptographic algorithms
  - SHA-1/256 hash module
  - True Random Number Generator (TRNG)
  - Real-time decryption for NOR Flash

| Product Part Numbers: |
| --- |
| HPM5361ICB1、HPM5361ICF1、HPM5361IEG1 |
| HPM5331ICB1、HPM5331ICF1、HPM5331IEG1 |
| HPM5321ICB1、HPM5321ICF1、HPM5321IEG1 |
| HPM5301IEG1 |

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# 1 Product Overview

## 1.1 System Block Diagram

The system block diagram of this product is shown in Figure 1.

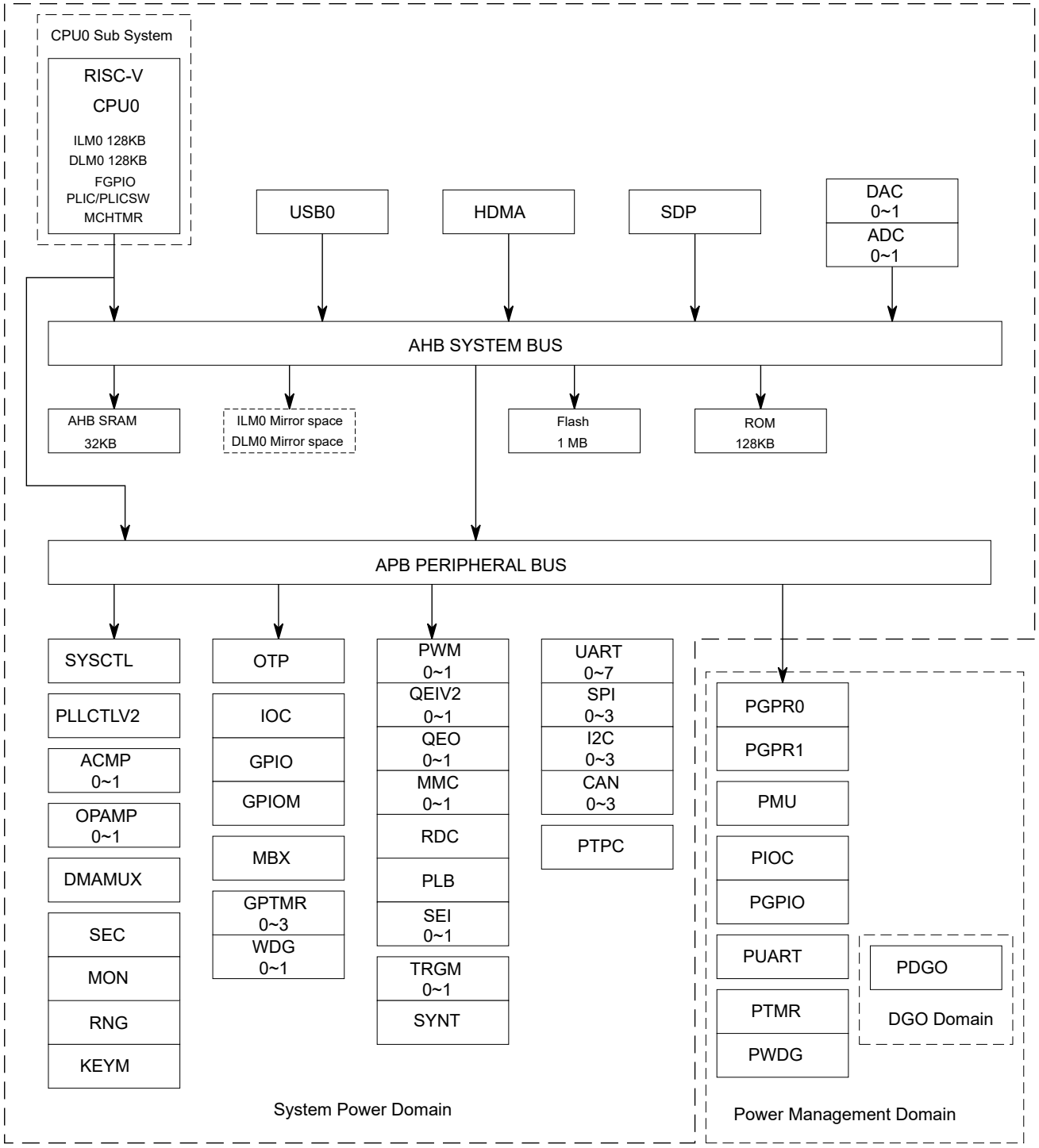



Figure 1: System Architecture Block Diagram

Table 1 summarizes the definitions of all peripheral abbreviations in Figure 1.

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| Abbreviation | Description |
|---|---|
| CPU0 Subsystem | Subsystem including RISC-V CPU0, its local memory and private peripherals |
| HART | Hardware Thread. As defined in the RISC-V specification, a HART is a unit that implements the complete RISC-V architecture and can execute instructions independently. In this manual, HART is equivalent to the RISC-V core. |
| ILM | Instruction Local Memory |
| DLM | Data Local Memory |
| FGPIO | Fast General Purpose Input Output Controller |
| USB | Universal Serial Bus |
| SDP | Secure Data Processor |
| HDMA | AHB Peripheral Bus DMA Controller |
| AHB SRAM | AHB Bus SRAM |
| EXIP | Encrypted Execution-In-Place Module |
| ADC | Analog-to-Digital Converter |
| DAC | Digital-to-Analog Converter |
| SYSCTL | System Control Module |
| PLLCTL | PLL Controller |
| ACMP | Analog Comparator |
| MBX | Mailbox |
| DMAMUX | DMA Request Router |
| IOC | Input Output Controller |
| PIOC | Power Management Domain IO Controller |
| GPIO | General Purpose Input Output Controller |
| PGPIO | Power Management Domain GPIO Controller |
| GPIOM | GPIO Manager |
| OTP | One Time Programmable Memory |
| PWM | Pulse Width Modulation Timer |
| QEIv2 | Quadrature Encoder Input |
| QEO | Quadrature Encoder Output |
| SEI | Serial Encoder Interface |
| MMC | Motion Management Controller |
| RDC | Resolver Decoder |
| PLB | Programmable Logic Block |
| TRGM | Trigger Manager |
| SYNT | Sync Timer |
| GPTMR | General Purpose Timer |
| PTMR | General Purpose Timer in Power Management Domain |
| EWDG | Watchdog |
| PWDG | Watchdog in Power Management Domain |

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| Abbreviation | Description |
|---|---|
| UART | Universal Asynchronous Receiver and Transmitter |
| PUART | Universal Asynchronous Receiver and Transmitter in Power Management Domain |
| SPI | Serial Peripheral Interface |
| I2C | Inter-Integrated Circuit Bus |
| CAN | Controller Area Network |
| PTPC | Precise Time Protocol Module |
| RNG | Random Number Generator |
| KEYM | Key Manager |
| PGPR0/1 | Power Management Domain General Purpose Registers 0/1 |
| PCFG | Power Management Domain Configuration Module |
| PDGO | Power Management Domain Power On/Off Module |
| SEC | Security Manager |
| MON | Security Monitor |
| System Power Domain | In this manual, system power domain specifically refers to the logic and memory circuits powered by VDD_SOC |
| Power Management Domain | In this manual, power management domain specifically refers to the logic and memory circuits powered by VPMC |

Table 1: Peripheral Abbreviation Summary

## 1.2 Feature Summary

This section describes the main features of this product.

### 1.2.1 Core and System

32-bit RISC-V processor with the following features:

- RV32-IMAFDCBP instruction set
  - Integer instruction set
  - Multiplication instruction set
  - Atomic instruction set
  - Single-precision floating-point instruction set
  - Double-precision floating-point instruction set
  - Compressed instruction set
  - Bit operation instruction set
  - DSP unit supporting SIMD and DSP instructions, compatible with P extension instruction set
- Performance up to 3.57 CoreMark / MHz
- Privilege modes support Machine mode and User mode
- Supports 16 Physical Memory Protection (PMP) regions
- Supports 16KB L1 instruction cache and 16KB L1 data cache

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- Supports 128 KB Instruction Local Memory (ILM) and 128 KB Data Local Memory (DLM)

The processor is equipped with 1 Platform Level Interrupt Controller (PLIC) for managing RISC-V external interrupts

- Supports multiple interrupt sources
- Supports 8 programmable interrupt priority levels
- Interrupt nesting extension and interrupt vector extension

The processor core is equipped with 1 software interrupt controller PLICSW to manage RISC-V software interrupts

- Generates RISC-V software interrupts

The processor core is equipped with 1 machine timer MCHTMR to manage RISC-V timer interrupts

- Generates RISC-V timer interrupts

DMA Controllers:

- HDMA, supporting 32 channels, for low-latency data transfer between peripheral registers and memory, also applicable for data transfer between memories
- Supports DMA request routing assigned to any DMA controller

Includes 1 Mailbox (MBX) supporting communication between different processes of the processor:

- Supports independent information sending and receiving interfaces
- Supports interrupt generation

### 1.2.2 Internal Memory

Internal memories include:

- 288 KB on-chip SRAM
  - ILM0, 128KB instruction local memory for RISC-V CPU0
  - DLM0, 128KB data local memory for RISC-V CPU0
  - AHB SRAM, 32KB, suitable for low-latency access by HDMA
- General purpose registers
  - Power management domain general purpose registers PGPR, 2 groups of 64 bytes each (128 bytes total), can retain data when the system power domain is powered off
  - DGO general purpose registers DGO_GPR, 16 bytes capacity, can retain data when the system power domain and power management domain are powered off
- Internal Read-Only Memory (ROM), 128KB capacity, stores the boot code of this product, Flashloader and part of peripheral driver programs
- One-Time Programmable memory (OTP), 4096 bits, can be used to store part of the chip's factory information, user keys, security configurations, boot configurations and other data

### 1.2.3 Power Management

This product integrates a complete power management system:

- Multiple on-chip power supplies
  - DCDC voltage converter, providing 0.9~1.3V output to power circuits in the system power domain. The DCDC output is adjustable to support Dynamic Voltage Frequency Scaling (DVFS)

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for optimizing power consumption during operation
  - LDOPMC, linear regulator with typical 1.1V output, powers circuits in the power management domain
  - LDOOTP, linear regulator with typical 2.5V output, powers OTP, can only be turned on during OTP programming
- Operation modes and low-power modes: Wait mode, Stop mode, Sleep mode and Shutdown mode
- The chip integrates power-on reset circuit
- The chip integrates low-voltage detection circuit

### 1.2.4 Clock

The clock management system of this product supports multiple clock sources and clock low-power management:

- External clock sources:
  - 24MHz on-chip oscillator OSC24M, supports 24MHz crystal, also supports external 24MHz active clock input through pins
- Internal clock sources:
  - Internal RC oscillator RC24M, 24MHz frequency, can be configured as alternate clock source for PLL
  - Internal 32KHz RC oscillator RC32K
- 2 Phase-Locked Loops (PLL), supporting fractional frequency division and spread spectrum
- Supports low-power management, supports automatic clock gating

### 1.2.5 Reset

Global reset can reset the entire chip including power management domain and system power domain. Reset sources include:

- RESETN pin reset (RESETN)

System power domain reset can reset the system power domain. Reset sources include:

- VPMC pin brownout reset (VPMC Brownout)
- Debug reset (DEBUG RST)
- Watchdog reset (WDOGx RST)
- Software reset (SW RST)

### 1.2.6 Boot

BootROM is the first program executed after the chip is powered on, it supports the following functions:

- Boot from serial NOR FLASH
- UART/USB boot
- In-System Programming (ISP)
- Secure boot
- Low-power wakeup
- Multiple ROM APIs

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### 1.2.7 External Memory

External memory interfaces include:

- 1 serial bus controller XPI, which can connect various off-chip SPI serial storage devices and serial bus compatible devices. Each XPI:
  - Supports 1/2/4-bit data mode, supports 2 CS chip select signals
  - Supports SDR and DDR, maximum 166MHz
  - Supports Quad-SPI and Octal-SPI serial NOR Flash
  - Supports serial NAND Flash
  - Supports HyperBus, HyperRAM and HyperFlash
  - Supports Quad/Oct SPI PSRAM

### 1.2.8 Motion Control System

Motion control system includes:

- 2 8-channel PWM timers, PWM modulation accuracy up to 3.0ns, supports complementary PWM output, dead time insertion and fault protection
- 2 Quadrature Encoder Input (QEIv2)
- 2 Quadrature Encoder Output (QEO)
- 2 Motion Management Controllers (MMC)
- 1 Resolver Decoder (RDC)
- 1 Programmable Logic Block (PLB)
- 2 Serial Encoder Interfaces (SEI)
- 1 Trigger Manager (TRGM), each module can interact with modules inside or outside the motion control system through TRGM
- 1 sync timer for synchronization

### 1.2.9 Timers

Timers include:

- 5 groups of 32-bit general purpose timers, one group (PTMR) is located in the power management domain, supports low-power wakeup. Each group of general purpose timers includes 4 32-bit counters
- 3 watchdogs, one (PWDG) is located in the power management domain

### 1.2.10 Communication Peripherals

Supports a rich set of communication peripherals, including:

- 9 Universal Asynchronous Receiver/Transmitters (UART), one (PUART) is located in the power management domain, supports low-power wakeup
- 4 Serial Peripheral Interfaces (SPI)
- 4 Inter-Integrated Circuit (I2C) buses, supporting Standard (100kbps), Fast (400kbps) and Fast+ (1 Mbps) modes
- 4 Controller Area Networks (CAN), supporting CAN_FD
  - Supports CAN 2.0B standard, 1Mbps
  - Supports CAN FD, 8 Mbps

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  - Supports timestamp
- 1 Precise Time Protocol Module (PTPC), supporting 2 groups of timestamp modules, each group includes a 64-bit counter, connected to the CAN module, the CAN module can read timestamp information from the port at any time
- 1 USB OTG controller, integrated with 1 high-speed USB-PHY
  - Complies with *Universal Serial Bus Specification Rev. 2.0*

### 1.2.11 Analog Peripherals

Analog peripherals include:

- 2 16-bit Analog-to-Digital Converters (ADC)
  - 16-bit successive approximation ADC
  - Supports 16 input channels
  - 2M sampling rate, 4M sampling rate when conversion accuracy is set to 12-bit
- 2 high-speed comparators
  - Operating voltage 3.0 ~ 3.6V, supports rail-to-rail input
  - Built-in 8-bit DAC
- 2 Digital-to-Analog Converters (DAC)
  - 12-bit accuracy, 1MSPS, supports output buffer
- 2 operational amplifiers (OPAMP)
  - Supports PGA mode

### 1.2.12 Input and Output

- Provides 8 groups from PA to PY with up to 56 GPIO function multiplexing pins
- IO supports 3V voltage mode, group power supply
- IO supports open-drain control, internal pull-up/down, drive strength adjustment, built-in Schmitt trigger
- GPIO controller
  - Supports reading input of any IO or controlling output of IO
  - Supports IO input trigger interrupt
- Fast GPIO controller (FGPIO), as a private IO fast access interface for the processor
- Provides a GPIO manager to manage IO control permissions of each GPIO controller
- Power management domain dedicated IO PYxx has exclusive GPIO controller and IO configuration module, supports state retention in low-power mode

### 1.2.13 Information Security System

Information security modules include:

- Secure Data Processor (SDP), the on-chip encryption and decryption algorithm engine:
  - Supports AES-128/256, SM4, supports ECB mode and CBC mode
  - Supports SHA-1/SHA-256, SM3
- Encrypted Execution-In-Place module (EXIP):
  - Tightly coupled with serial bus controller XPI, supports online decryption of external NOR Flash
  - AES-128 CTR mode, zero-wait cycle decryption
  - Supports RFC3394 key unwrapping, protects Data Encryption Key (DEK) through Key Encryp-

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tion Key (KEK)

- Key Manager (KEYM):
  - Supports loading keys from OTP key area through independent data path
  - Supports key obfuscation
  - Supports loading random keys from True Random Number Generator (RNG)
  - Supports generating Session Key
  - Supports independent data path to transmit keys to Secure Data Processor (SDP)
- Key area in OTP, supports storage and protection of:
  - Related keys for SDP and EXIP
  - Related keys for secure boot
  - Related keys for secure debugging
  - Product life cycle configuration
- True Random Number Generator (RNG):
  - 3 independent entropy sources are internal analog noise sources
- Security Manager (SEC):
  - Monitors product life cycle
  - Configures system security status
  - Formulates security rules and monitors security rule violation events
  - Associated with Monitor (MON), monitors VPMC power supply and OSC24M clock
- Secure boot mechanism based on BOOT ROM, supports encrypted boot, supports trusted execution environment

### 1.2.14 System Debugging

System debugging modules include:

- Supports JTAG interface
  - Complies with *RISC-V External Debug Support V0.13* specification
  - Supports IEEE1149.1
  - Accesses RISC-V core registers, CSRs and memory
- Debug port locking function
  - Open mode: debugging function is enabled
  - Locked mode: debugging function is disabled, can be unlocked with debug key
  - Disabled mode: debugging function is permanently disabled

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# 2 Pin and Function Description

## 2.1 LQFP100 Pin Distribution

The LQFP100 pin distribution (top view) is shown in Figure 2.

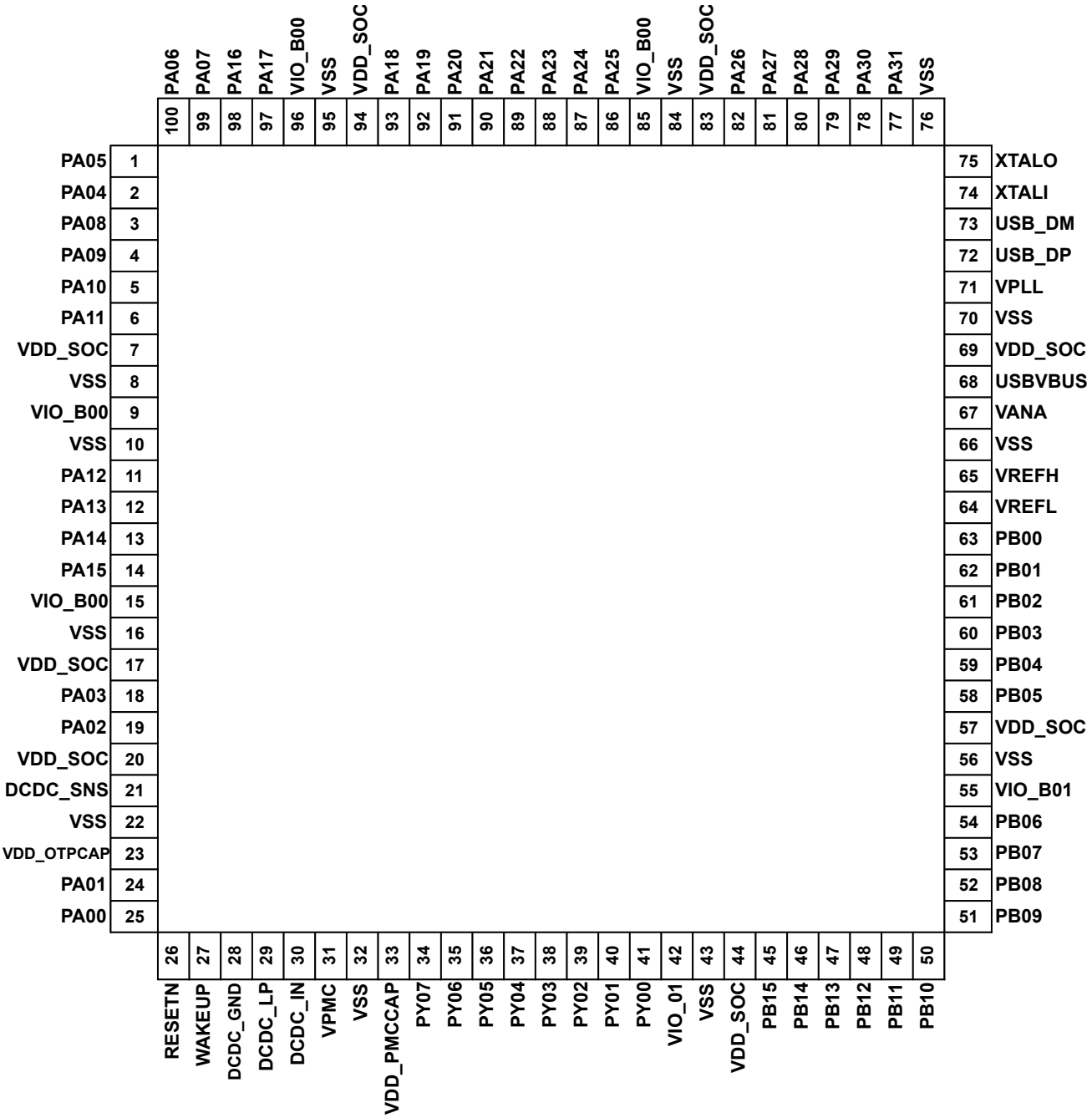



Figure 2: LQFP100 Pin Distribution

## 2.2 LQFP64 Pin Distribution

The LQFP64 pin distribution (top view) is shown in Figure 3.

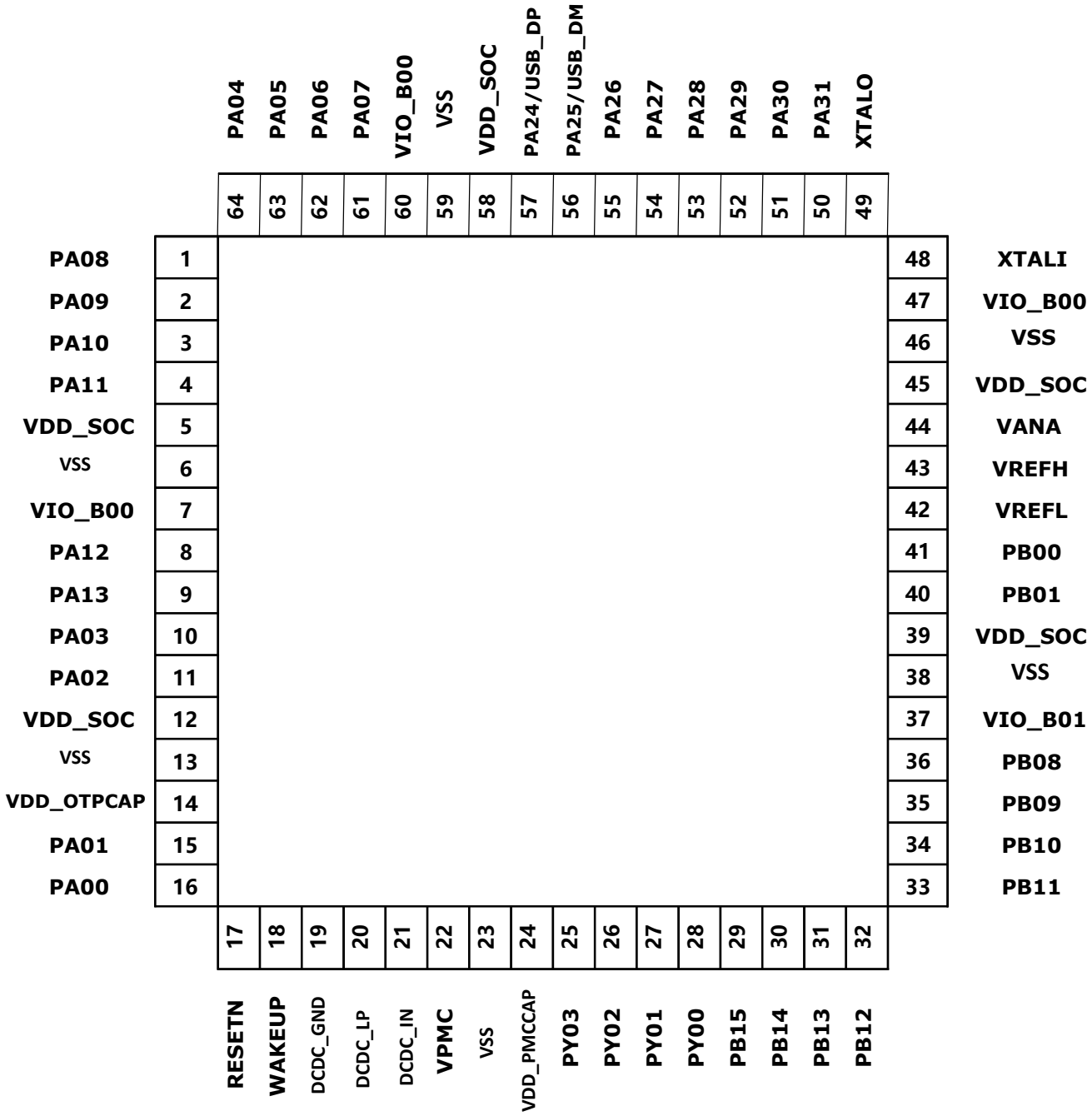



Figure 3: LQFP64 Pin Distribution

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## 2.3 QFN48 Pin Distribution

The QFN48 pin distribution (top view) is shown in Figure 4.

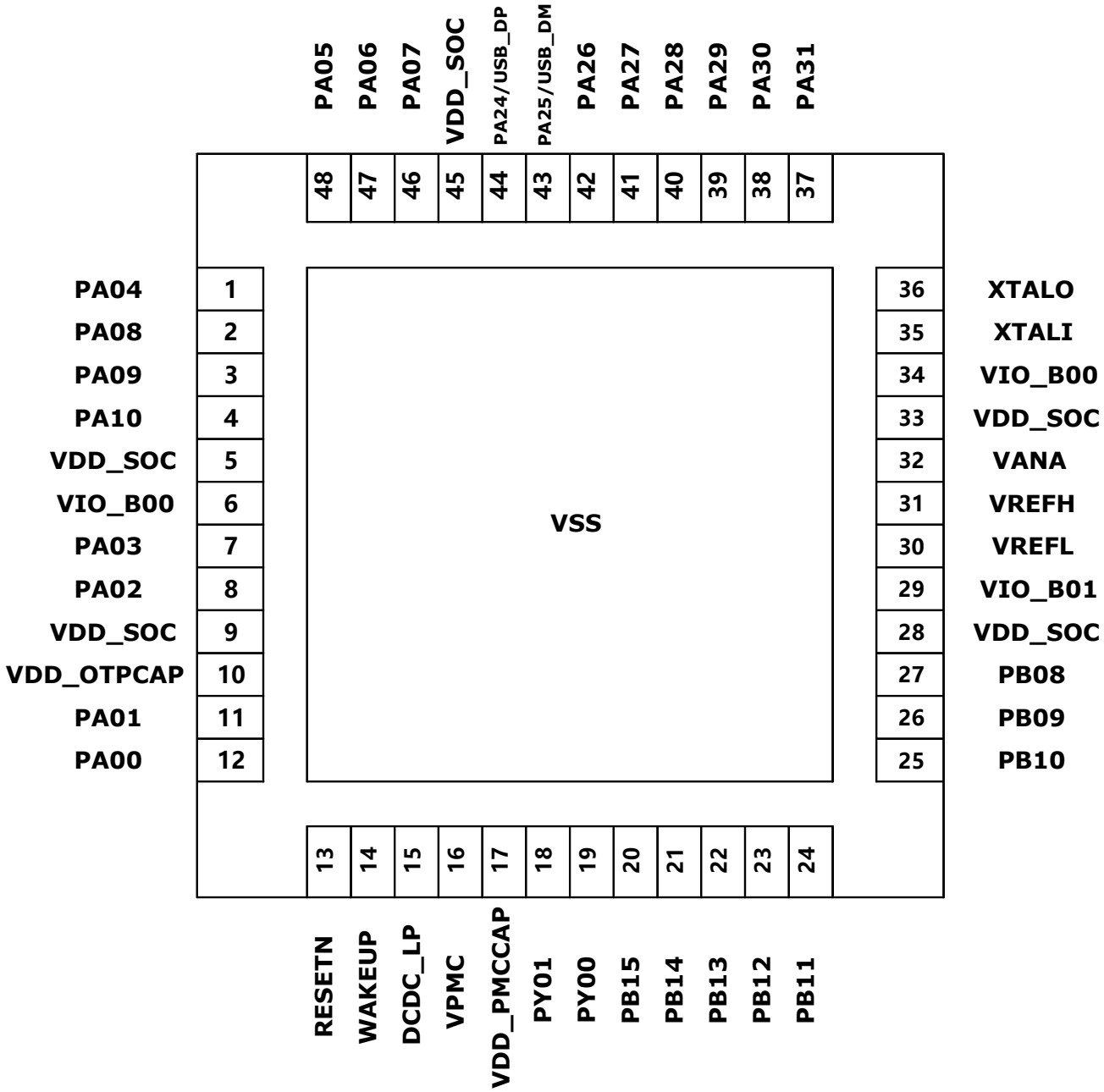



Figure 4: QFN48 Pin Distribution

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## 2.4 Pin Configuration and Pin Multiplexing (PINMUX)

The pin configuration and functions of the HPM5300 series are as follows:

| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 25 | 20 | 16 | 12 | PA00 | GPIO_A_00(ALT0)<br>GPTMR1_COMP_0(ALT1)<br>UART0_TXD(ALT2)<br>MCAN0_TXD(ALT7)<br>PWM0_FAULT_0(ALT16)<br>PWM1_P_0(ALT17)<br>TRGM0_P_00(ALT18)<br>PWM1_FAULT_0(ALT19)<br>SYSCTL_CLK_OBS_0(ALT24) | - | B00 |
| 24 | 19 | 15 | 11 | PA01 | GPIO_A_01(ALT0)<br>GPTMR1_CAPT_0(ALT1)<br>UART0_RXD(ALT2)<br>MCAN0_RXD(ALT7)<br>PWM0_FAULT_1(ALT16)<br>PWM1_P_1(ALT17)<br>TRGM0_P_01(ALT18)<br>ACMP_COMP_0(ALT19)<br>SYSCTL_CLK_OBS_1(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 19 | 14 | 11 | 8 | PA02 | GPIO_A_02(ALT0)<br>GPTMR1_COMP_1(ALT1)<br>UART0_DE(ALT2)<br>UART0_RTS(ALT3)<br>I2C0_SCL(ALT4)<br>MCAN0_STBY(ALT7)<br>ACMP_COMP_0(ALT16)<br>PWM1_P_2(ALT17)<br>TRGM0_P_02(ALT18)<br>ACMP_COMP_1(ALT19)<br>QEI1_F(ALT20)<br>SYSCTL_CLK_OBS_2(ALT24) | - | B00 |
| 18 | 13 | 10 | 7 | PA03 | GPIO_A_03(ALT0)<br>GPTMR1_CAPT_1(ALT1)<br>UART0_CTS(ALT3)<br>I2C0_SDA(ALT4)<br>SPI3_CS_3(ALT5)<br>MCAN1_STBY(ALT7)<br>ACMP_COMP_1(ALT16)<br>PWM1_P_3(ALT17)<br>TRGM0_P_03(ALT18)<br>PWM1_FAULT_1(ALT19)<br>QEI1_H1(ALT20)<br>SYSCTL_CLK_OBS_3(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 2 | 80 | 64 | 1 | PA04 | GPIO_A_04(ALT0)<br>UART1_CTS(ALT3)<br>SPI0_CS_0(ALT5)<br>MCAN1_RXD(ALT7)<br>PWM0_P_0(ALT16)<br>PWM1_P_4(ALT17)<br>TRGM0_P_04(ALT18)<br>RDC0_EXC_P(ALT19)<br>QEI1_A(ALT20)<br>QEO1_A(ALT21)<br>SEI1_DE(ALT22)<br>JTAG_TDO(ALT24) | - | B00 |
| 1 | 79 | 63 | 48 | PA05 | GPIO_A_05(ALT0)<br>GPTMR1_COMP_2(ALT1)<br>UART1_DE(ALT2)<br>UART1_RTS(ALT3)<br>SPI0_SCLK(ALT5)<br>MCAN1_TXD(ALT7)<br>PWM0_P_1(ALT16)<br>PWM1_P_5(ALT17)<br>TRGM0_P_05(ALT18)<br>RDC0_EXC_N(ALT19)<br>QEI1_B(ALT20)<br>QEO1_B(ALT21)<br>SEI1_CK(ALT22)<br>JTAG_TDI(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 100 | 78 | 62 | 47 | PA06 | GPIO_A_06(ALT0)<br>GPTMR0_CAPT_0(ALT1)<br>UART1_RXD(ALT2)<br>I2C1_SDA(ALT4)<br>SPI0_MISO(ALT5)<br>PWM0_P_2(ALT16)<br>PWM1_P_6(ALT17)<br>TRGM0_P_06(ALT18)<br>QEI1_Z(ALT20)<br>QEO1_Z(ALT21)<br>SEI1_TX(ALT22)<br>JTAG_TCK(ALT24) | - | B00 |
| 99 | 77 | 61 | 46 | PA07 | GPIO_A_07(ALT0)<br>GPTMR0_COMP_0(ALT1)<br>UART1_TXD(ALT2)<br>I2C1_SCL(ALT4)<br>SPI0_MOSI(ALT5)<br>PWM0_P_3(ALT16)<br>PWM1_P_7(ALT17)<br>TRGM0_P_07(ALT18)<br>QEI1_H0(ALT20)<br>SEI1_RX(ALT22)<br>JTAG_TMS(ALT24) | - | B00 |
| 3 | 1 | 1 | 2 | PA08 | GPIO_A_08(ALT0)<br>GPTMR0_COMP_1(ALT1)<br>UART2_TXD(ALT2)<br>I2C2_SCL(ALT4)<br>SPI3_CS_2(ALT5)<br>MCAN2_TXD(ALT7)<br>PWM0_P_4(ALT16)<br>PWM0_FAULT_0(ALT18)<br>JTAG_TRST(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 4 | 2 | 2 | 3 | PA09 | GPIO_A_09(ALT0)<br>GPTMR0_CAPT_1(ALT1)<br>UART2_RXD(ALT2)<br>I2C2_SDA(ALT4)<br>SPI3_CS_1(ALT5)<br>MCAN2_RXD(ALT7)<br>PWM0_P_5(ALT16)<br>PWM0_FAULT_1(ALT18)<br>SOC_REF0(ALT24) | - | B00 |
| 5 | 3 | 3 | 4 | PA10 | GPIO_A_10(ALT0)<br>GPTMR0_COMP_2(ALT1)<br>UART2_DE(ALT2)<br>UART2_RTS(ALT3)<br>SPI3_CS_0(ALT5)<br>MCAN2_STBY(ALT7)<br>PWM0_P_6(ALT16)<br>PWM1_FAULT_0(ALT17)<br>ACMP_COMP_0(ALT18)<br>QEI1_A(ALT20)<br>QEO0_A(ALT21)<br>SEI1_DE(ALT22) | - | B00 |
| 6 | 4 | 4 | - | PA11 | GPIO_A_11(ALT0)<br>UART2_CTS(ALT3)<br>SPI3_SCLK(ALT5)<br>PWM0_P_7(ALT16)<br>PWM1_FAULT_1(ALT17)<br>ACMP_COMP_1(ALT18)<br>QEI1_B(ALT20)<br>QEO0_B(ALT21)<br>SEI1_CK(ALT22)<br>EWDG0_RST(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 11 | 7 | 8 | - | PA12 | GPIO_A_12(ALT0)<br>UART3_CTS(ALT3)<br>I2C3_SDA(ALT4)<br>SPI3_MISO(ALT5)<br>PWM0_P_0(ALT16)<br>PWM1_FAULT_0(ALT17)<br>PWM0_FAULT_0(ALT18)<br>RDC0_EXC_P(ALT19)<br>QEI1_Z(ALT20)<br>QEO0_Z(ALT21)<br>SEI1_TX(ALT22) | - | B00 |
| 12 | 8 | 9 | - | PA13 | GPIO_A_13(ALT0)<br>GPTMR1_COMP_3(ALT1)<br>UART3_DE(ALT2)<br>UART3_RTS(ALT3)<br>I2C3_SCL(ALT4)<br>SPI3_MOSI(ALT5)<br>MCAN3_STBY(ALT7)<br>PWM0_P_1(ALT16)<br>PWM1_FAULT_1(ALT17)<br>PWM0_FAULT_1(ALT18)<br>RDC0_EXC_N(ALT19)<br>QEI1_H0(ALT20)<br>SEI1_RX(ALT22) | - | B00 |
| 13 | 9 | - | - | PA14 | GPIO_A_14(ALT0)<br>UART3_RXD(ALT2)<br>SPI3_DAT2(ALT5)<br>MCAN3_RXD(ALT7)<br>PWM0_P_2(ALT16)<br>ACMP_COMP_0(ALT18)<br>QEI1_H1(ALT20)<br>EWDG1_RST(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 14 | 10 | - | - | PA15 | GPIO_A_15(ALT0)<br>GPTMR0_COMP_3(ALT1)<br>UART3_TXD(ALT2)<br>SPI3_DAT3(ALT5)<br>MCAN3_TXD(ALT7)<br>PWM0_P_3(ALT16)<br>ACMP_COMP_1(ALT18)<br>QEI1_F(ALT20)<br>SOC_REF0(ALT24) | - | B00 |
| 98 | 76 | - | - | PA16 | GPIO_A_16(ALT0)<br>GPTMR3_COMP_0(ALT1)<br>UART4_TXD(ALT2)<br>MCAN0_TXD(ALT7)<br>PWM0_P_4(ALT16)<br>PWM1_P_0(ALT17)<br>TRGM0_P_04(ALT18)<br>QEO0_A(ALT21)<br>SEI1_DE(ALT22) | - | B00 |
| 97 | 75 | - | - | PA17 | GPIO_A_17(ALT0)<br>GPTMR3_CAPT_0(ALT1)<br>UART4_RXD(ALT2)<br>MCAN0_RXD(ALT7)<br>PWM0_P_5(ALT16)<br>PWM1_P_1(ALT17)<br>TRGM0_P_05(ALT18)<br>QEO0_B(ALT21)<br>SEI1_CK(ALT22) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 93 | 72 | - | - | PA18 | GPIO_A_18(ALT0)<br>GPTMR3_COMP_1(ALT1)<br>UART4_DE(ALT2)<br>UART4_RTS(ALT3)<br>I2C0_SCL(ALT4)<br>MCAN0_STBY(ALT7)<br>PWM0_P_6(ALT16)<br>PWM1_P_2(ALT17)<br>TRGM0_P_06(ALT18)<br>QEO0_Z(ALT21)<br>SEI1_TX(ALT22) | - | B00 |
| 92 | 71 | - | - | PA19 | GPIO_A_19(ALT0)<br>GPTMR3_CAPT_1(ALT1)<br>UART4_CTS(ALT3)<br>I2C0_SDA(ALT4)<br>SPI1_CS_3(ALT5)<br>MCAN1_STBY(ALT7)<br>PWM0_P_7(ALT16)<br>PWM1_P_3(ALT17)<br>TRGM0_P_07(ALT18)<br>SEI1_RX(ALT22) | - | B00 |
| 91 | - | - | - | PA20 | GPIO_A_20(ALT0)<br>UART5_CTS(ALT3)<br>SPI2_CS_0(ALT5)<br>MCAN1_RXD(ALT7)<br>PWM0_FAULT_0(ALT16)<br>PWM1_P_4(ALT17)<br>TRGM0_P_00(ALT18)<br>QEI0_A(ALT20)<br>QEO0_A(ALT21)<br>SEI0_DE(ALT22) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 90 | - | - | - | PA21 | GPIO_A_21(ALT0)<br>GPTMR3_COMP_2(ALT1)<br>UART5_DE(ALT2)<br>UART5_RTS(ALT3)<br>SPI2_SCLK(ALT5)<br>MCAN1_TXD(ALT7)<br>PWM0_FAULT_1(ALT16)<br>PWM1_P_5(ALT17)<br>TRGM0_P_01(ALT18)<br>QEI0_B(ALT20)<br>QEO0_B(ALT21)<br>SEI0_CK(ALT22) | - | B00 |
| 89 | - | - | - | PA22 | GPIO_A_22(ALT0)<br>GPTMR2_CAPT_0(ALT1)<br>UART5_RXD(ALT2)<br>I2C1_SDA(ALT4)<br>SPI2_MISO(ALT5)<br>PWM1_P_6(ALT17)<br>TRGM0_P_02(ALT18)<br>PWM1_FAULT_0(ALT19)<br>QEI0_Z(ALT20)<br>QEO0_Z(ALT21)<br>SEI0_TX(ALT22) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 88 | - | - | - | PA23 | GPIO_A_23(ALT0)<br>GPTMR2_COMP_0(ALT1)<br>UART5_TXD(ALT2)<br>I2C1_SCL(ALT4)<br>SPI2_MOSI(ALT5)<br>PWM1_P_7(ALT17)<br>TRGM0_P_03(ALT18)<br>PWM1_FAULT_1(ALT19)<br>QEI0_H0(ALT20)<br>SEI0_RX(ALT22) | - | B00 |
| 87 | 70 | 57 | 44 | PA24 | GPIO_A_24(ALT0)<br>GPTMR2_COMP_1(ALT1)<br>UART6_TXD(ALT2)<br>I2C2_SCL(ALT4)<br>SPI1_CS_2(ALT5)<br>MCAN2_TXD(ALT7)<br>XPI0_CA_CS1(ALT14)<br>PWM0_P_0(ALT16)<br>PWM1_P_0(ALT17)<br>TRGM0_P_00(ALT18)<br>QEI0_H1(ALT20) | - | B00 |
| 86 | 69 | 56 | 43 | PA25 | GPIO_A_25(ALT0)<br>GPTMR2_CAPT_1(ALT1)<br>UART6_RXD(ALT2)<br>I2C2_SDA(ALT4)<br>SPI1_CS_1(ALT5)<br>MCAN2_RXD(ALT7)<br>XPI0_CA_DQS(ALT14)<br>PWM0_P_1(ALT16)<br>PWM1_P_1(ALT17)<br>TRGM0_P_01(ALT18)<br>QEI0_F(ALT20) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 82 | 66 | 55 | 42 | PA26 | GPIO_A_26(ALT0)<br>GPTMR2_COMP_2(ALT1)<br>UART6_DE(ALT2)<br>UART6_RTS(ALT3)<br>SPI1_CS_0(ALT5)<br>MCAN2_STBY(ALT7)<br>XPI0_CA_D_3(ALT14)<br>PWM0_P_2(ALT16)<br>PWM1_P_2(ALT17)<br>TRGM0_P_02(ALT18)<br>QEI0_A(ALT20)<br>QEO0_A(ALT21)<br>SEI0_DE(ALT22)<br>SYSCTL_CLK_OBS_0(ALT24) | - | B00 |
| 81 | 65 | 54 | 41 | PA27 | GPIO_A_27(ALT0)<br>UART6_CTS(ALT3)<br>SPI1_SCLK(ALT5)<br>XPI0_CA_SCLK(ALT14)<br>PWM0_P_3(ALT16)<br>PWM1_P_3(ALT17)<br>TRGM0_P_03(ALT18)<br>QEI0_B(ALT20)<br>QEO0_B(ALT21)<br>SEI0_CK(ALT22)<br>SYSCTL_CLK_OBS_1(ALT24) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 80 | 64 | 53 | 40 | PA28 | GPIO_A_28(ALT0)<br>UART7_CTS(ALT3)<br>I2C3_SDA(ALT4)<br>SPI1_MISO(ALT5)<br>XPI0_CA_D_0(ALT14)<br>PWM0_P_4(ALT16)<br>PWM1_P_4(ALT17)<br>TRGM0_P_04(ALT18)<br>RDC0_EXC_P(ALT19)<br>QEI0_Z(ALT20)<br>QEO0_Z(ALT21)<br>SEI0_TX(ALT22)<br>SYSCTL_CLK_OBS_2(ALT24) | - | B00 |
| 79 | 63 | 52 | 39 | PA29 | GPIO_A_29(ALT0)<br>GPTMR3_COMP_3(ALT1)<br>UART7_DE(ALT2)<br>UART7_RTS(ALT3)<br>I2C3_SCL(ALT4)<br>SPI1_MOSI(ALT5)<br>MCAN3_STBY(ALT7)<br>XPI0_CA_D_2(ALT14)<br>PWM0_P_5(ALT16)<br>PWM1_P_5(ALT17)<br>TRGM0_P_05(ALT18)<br>RDC0_EXC_N(ALT19)<br>QEI0_H0(ALT20)<br>SEI0_RX(ALT22)<br>SYSCTL_CLK_OBS_3(ALT24)<br>USB0_OC(ALT25) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 78 | 62 | 51 | 38 | PA30 | GPIO_A_30(ALT0)<br>UART7_RXD(ALT2)<br>SPI1_DAT2(ALT5)<br>MCAN3_RXD(ALT7)<br>XPI0_CA_D_1(ALT14)<br>PWM0_P_6(ALT16)<br>PWM1_P_6(ALT17)<br>TRGM0_P_06(ALT18)<br>QEI0_H1(ALT20)<br>SOC_REF0(ALT24)<br>USB0_PWR(ALT25) | - | B00 |
| 77 | 61 | 50 | 37 | PA31 | GPIO_A_31(ALT0)<br>GPTMR2_COMP_3(ALT1)<br>UART7_TXD(ALT2)<br>SPI1_DAT3(ALT5)<br>MCAN3_TXD(ALT7)<br>XPI0_CA_CS0(ALT14)<br>PWM0_P_7(ALT16)<br>PWM1_P_7(ALT17)<br>TRGM0_P_07(ALT18)<br>QEI0_F(ALT20)<br>USB0_ID(ALT25) | - | B00 |
| 63 | 49 | 41 | - | PB00 | GPIO_B_00(ALT0)<br>GPTMR1_COMP_0(ALT1)<br>UART0_TXD(ALT2)<br>MCAN0_TXD(ALT7)<br>PWM0_P_0(ALT16)<br>PWM1_FAULT_0(ALT17)<br>TRGM0_P_04(ALT18)<br>ACMP_COMP_0(ALT19) | ADC0_IN15<br>ADC1_IN15<br>ACMP_CMP0_INN7<br>OPA0_OUT | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 62 | 48 | 40 | - | PB01 | GPIO_B_01(ALT0)<br>GPTMR1_CAPT_0(ALT1)<br>UART0_RXD(ALT2)<br>MCAN0_RXD(ALT7)<br>PWM0_P_1(ALT16)<br>PWM1_FAULT_1(ALT17)<br>TRGM0_P_05(ALT18)<br>ACMP_COMP_1(ALT19) | ADC0_IN14<br>ADC1_IN14<br>ACMP_CMP1_INN7<br>OPA1_OUT | B01 |
| 61 | - | - | - | PB02 | GPIO_B_02(ALT0)<br>GPTMR1_COMP_1(ALT1)<br>UART0_DE(ALT2)<br>UART0_RTS(ALT3)<br>I2C0_SCL(ALT4)<br>MCAN0_STBY(ALT7)<br>PWM0_P_2(ALT16)<br>ACMP_COMP_1(ALT17)<br>TRGM0_P_06(ALT18)<br>PWM0_FAULT_0(ALT19) | ADC0_IN12<br>ADC1_IN12<br>ACMP_CMP0_INP7<br>OPA0_EXT | B01 |
| 60 | - | - | - | PB03 | GPIO_B_03(ALT0)<br>GPTMR1_CAPT_1(ALT1)<br>UART0_CTS(ALT3)<br>I2C0_SDA(ALT4)<br>SPI2_CS_3(ALT5)<br>MCAN1_STBY(ALT7)<br>PWM0_P_3(ALT16)<br>ACMP_COMP_0(ALT17)<br>TRGM0_P_07(ALT18)<br>PWM0_FAULT_1(ALT19) | ADC0_IN8<br>ADC1_IN8<br>ACMP_CMP1_INP7<br>OPA1_EXT | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 59 | - | - | - | PB04 | GPIO_B_04(ALT0)<br>UART1_CTS(ALT3)<br>SPI3_CS_0(ALT5)<br>MCAN1_RXD(ALT7)<br>PWM0_P_4(ALT16)<br>PWM1_P_0(ALT17)<br>TRGM0_P_00(ALT18)<br>QEI1_A(ALT20)<br>QEO1_A(ALT21)<br>SEI0_DE(ALT22) | ADC0_IN0<br>ADC1_IN0<br>ACMP_CMP0_INN5<br>ACMP_CMP1_INN5<br>OPA0_INP0 | B01 |
| 58 | - | - | - | PB05 | GPIO_B_05(ALT0)<br>GPTMR1_COMP_2(ALT1)<br>UART1_DE(ALT2)<br>UART1_RTS(ALT3)<br>SPI3_SCLK(ALT5)<br>MCAN1_TXD(ALT7)<br>PWM0_P_5(ALT16)<br>PWM1_P_1(ALT17)<br>TRGM0_P_01(ALT18)<br>QEI1_B(ALT20)<br>QEO1_B(ALT21)<br>SEI0_CK(ALT22) | ADC0_IN13<br>ADC1_IN13<br>ACMP_CMP0_INN3<br>ACMP_CMP1_INN3<br>OPA0_INN0 | B01 |
| 54 | - | - | - | PB06 | GPIO_B_06(ALT0)<br>GPTMR0_CAPT_0(ALT1)<br>UART1_RXD(ALT2)<br>I2C1_SDA(ALT4)<br>SPI3_MISO(ALT5)<br>PWM0_P_6(ALT16)<br>PWM1_P_2(ALT17)<br>TRGM0_P_02(ALT18)<br>RDC0_EXC_P(ALT19)<br>QEI1_Z(ALT20)<br>QEO1_Z(ALT21)<br>SEI0_TX(ALT22) | ADC0_IN9<br>ADC1_IN9<br>ACMP_CMP0_INP5<br>ACMP_CMP1_INP5<br>OPA0_INP1 | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 53 | - | - | - | PB07 | GPIO_B_07(ALT0)<br>GPTMR0_COMP_0(ALT1)<br>UART1_TXD(ALT2)<br>I2C1_SCL(ALT4)<br>SPI3_MOSI(ALT5)<br>PWM0_P_7(ALT16)<br>PWM1_P_3(ALT17)<br>TRGM0_P_03(ALT18)<br>RDC0_EXC_N(ALT19)<br>QEI1_H0(ALT20)<br>SEI0_RX(ALT22) | ADC0_IN10<br>ADC1_IN10<br>ACMP_CMP0_INP3<br>ACMP_CMP1_INP3<br>OPA0_INN1 | B01 |
| 52 | 44 | 36 | 27 | PB08 | GPIO_B_08(ALT0)<br>GPTMR0_COMP_1(ALT1)<br>UART2_TXD(ALT2)<br>I2C2_SCL(ALT4)<br>SPI2_CS_2(ALT5)<br>MCAN2_TXD(ALT7)<br>ACMP_COMP_0(ALT16)<br>PWM1_P_4(ALT17)<br>QEI1_H1(ALT20)<br>QEO1_A(ALT21)<br>SEI1_DE(ALT22)<br>USB0_ID(ALT25) | ADC0_IN11<br>ADC1_IN11<br>DAC0_OUT<br>ACMP_CMP0_INN6<br>ACMP_CMP1_INN6<br>OPA0_INP2<br>OPA1_INP2 | B01 |
| 51 | 43 | 35 | 26 | PB09 | GPIO_B_09(ALT0)<br>GPTMR0_CAPT_1(ALT1)<br>UART2_RXD(ALT2)<br>I2C2_SDA(ALT4)<br>SPI2_CS_1(ALT5)<br>MCAN2_RXD(ALT7)<br>ACMP_COMP_1(ALT16)<br>PWM1_P_5(ALT17)<br>QEI1_F(ALT20)<br>QEO1_B(ALT21)<br>SEI1_CK(ALT22)<br>USB0_OC(ALT25) | ADC0_IN1<br>ADC1_IN1<br>DAC1_OUT<br>ACMP_CMP0_INP6<br>ACMP_CMP1_INP6<br>OPA0_INN2<br>OPA1_INN2 | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 50 | 42 | 34 | 25 | PB10 | GPIO_B_10(ALT0)<br>GPTMR0_COMP_2(ALT1)<br>UART2_DE(ALT2)<br>UART2_RTS(ALT3)<br>SPI2_CS_0(ALT5)<br>MCAN2_STBY(ALT7)<br>ACMP_COMP_0(ALT16)<br>PWM1_P_6(ALT17)<br>QEI0_H1(ALT20)<br>QEO1_Z(ALT21)<br>SEI1_TX(ALT22)<br>USB0_PWR(ALT25) | ADC0_IN2<br>ADC1_IN2<br>ACMP_CMP0_INP4<br>ACMP_CMP1_INP4<br>OPA0_INP3<br>OPA1_INP3 | B01 |
| 49 | 41 | 33 | 24 | PB11 | GPIO_B_11(ALT0)<br>UART2_CTS(ALT3)<br>SPI2_SCLK(ALT5)<br>ACMP_COMP_1(ALT16)<br>PWM1_P_7(ALT17)<br>QEI0_F(ALT20)<br>SEI1_RX(ALT22) | ADC0_IN3<br>ADC1_IN3<br>ACMP_CMP0_INN4<br>ACMP_CMP1_INN4<br>OPA0_INN3<br>OPA1_INN3 | B01 |
| 48 | 40 | 32 | 23 | PB12 | GPIO_B_12(ALT0)<br>UART3_CTS(ALT3)<br>I2C3_SDA(ALT4)<br>SPI2_MISO(ALT5)<br>PWM1_FAULT_0(ALT16)<br>PWM1_P_0(ALT17)<br>TRGM0_P_00(ALT18)<br>QEI0_A(ALT20)<br>QEO1_A(ALT21)<br>SEI0_DE(ALT22) | ADC0_IN4<br>ADC1_IN4<br>ACMP_CMP0_INN2<br>ACMP_CMP1_INN2<br>OPA1_INP0 | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 47 | 39 | 31 | 22 | PB13 | GPIO_B_13(ALT0)<br>GPTMR1_COMP_3(ALT1)<br>UART3_DE(ALT2)<br>UART3_RTS(ALT3)<br>I2C3_SCL(ALT4)<br>SPI2_MOSI(ALT5)<br>MCAN3_STBY(ALT7)<br>PWM1_FAULT_1(ALT16)<br>PWM1_P_1(ALT17)<br>TRGM0_P_01(ALT18)<br>QEI0_B(ALT20)<br>QEO1_B(ALT21)<br>SEI0_CK(ALT22) | ADC0_IN5<br>ADC1_IN5<br>ACMP_CMP0_INP2<br>ACMP_CMP1_INP2<br>OPA1_INN0 | B01 |
| 46 | 38 | 30 | 21 | PB14 | GPIO_B_14(ALT0)<br>UART3_RXD(ALT2)<br>SPI2_DAT2(ALT5)<br>MCAN3_RXD(ALT7)<br>PWM0_FAULT_0(ALT16)<br>PWM1_P_2(ALT17)<br>TRGM0_P_02(ALT18)<br>RDC0_EXC_P(ALT19)<br>QEI0_Z(ALT20)<br>QEO1_Z(ALT21)<br>SEI0_TX(ALT22) | ADC0_IN6<br>ADC1_IN6<br>ACMP_CMP0_INN1<br>ACMP_CMP1_INN1<br>OPA1_INP1 | B01 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 45 | 37 | 29 | 20 | PB15 | GPIO_B_15(ALT0)<br>GPTMR0_COMP_3(ALT1)<br>UART3_TXD(ALT2)<br>SPI2_DAT3(ALT5)<br>MCAN3_TXD(ALT7)<br>PWM0_FAULT_1(ALT16)<br>PWM1_P_3(ALT17)<br>TRGM0_P_03(ALT18)<br>RDC0_EXC_N(ALT19)<br>QEI0_H0(ALT20)<br>SEI0_RX(ALT22) | ADC0_IN7<br>ADC1_IN7<br>ACMP_CMP0_INP1<br>ACMP_CMP1_INP1<br>OPA1_INN1 | B01 |
| - | - | - | - | PX00 | GPIO_X_00(ALT0)<br>GPTMR2_COMP_0(ALT1)<br>UART4_TXD(ALT2)<br>MCAN0_TXD(ALT7)<br>XPI0_CA_D_2(ALT14) | - | B00 |
| - | - | - | - | PX01 | GPIO_X_01(ALT0)<br>GPTMR2_CAPT_0(ALT1)<br>UART4_RXD(ALT2)<br>MCAN0_RXD(ALT7)<br>XPI0_CA_D_1(ALT14) | - | B00 |
| - | - | - | - | PX02 | GPIO_X_02(ALT0)<br>GPTMR2_COMP_1(ALT1)<br>UART4_DE(ALT2)<br>UART4_RTS(ALT3)<br>I2C0_SCL(ALT4)<br>MCAN0_STBY(ALT7)<br>XPI0_CA_CS0(ALT14) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| - | - | - | - | PX03 | GPIO_X_03(ALT0)<br>GPTMR2_CAPT_1(ALT1)<br>UART4_CTS(ALT3)<br>I2C0_SDA(ALT4)<br>MCAN1_STBY(ALT7)<br>XPI0_CA_DQS(ALT14) | - | B00 |
| - | - | - | - | PX04 | GPIO_X_04(ALT0)<br>UART5_CTS(ALT3)<br>SPI1_CS_0(ALT5)<br>MCAN1_RXD(ALT7)<br>XPI0_CA_CS1(ALT14) | - | B00 |
| - | - | - | - | PX05 | GPIO_X_05(ALT0)<br>GPTMR2_COMP_2(ALT1)<br>UART5_DE(ALT2)<br>UART5_RTS(ALT3)<br>SPI1_SCLK(ALT5)<br>MCAN1_TXD(ALT7)<br>XPI0_CA_D_0(ALT14) | - | B00 |
| - | - | - | - | PX06 | GPIO_X_06(ALT0)<br>GPTMR3_CAPT_0(ALT1)<br>UART5_RXD(ALT2)<br>I2C1_SDA(ALT4)<br>SPI1_MISO(ALT5)<br>XPI0_CA_SCLK(ALT14) | - | B00 |
| - | - | - | - | PX07 | GPIO_X_07(ALT0)<br>GPTMR3_COMP_0(ALT1)<br>UART5_TXD(ALT2)<br>I2C1_SCL(ALT4)<br>SPI1_MOSI(ALT5)<br>XPI0_CA_D_3(ALT14) | - | B00 |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 41 | 34 | 28 | 19 | PY00 | GPIO_Y_00(ALT0)<br>GPTMR3_COMP_0(ALT1)<br>UART0_TXD(ALT2)<br>MCAN2_TXD(ALT7)<br>PWM0_P_0(ALT16)<br>PWM1_P_4(ALT17)<br>PWM0_FAULT_0(ALT18)<br>USB0_ID(ALT25) | - | VPMC |
| 40 | 33 | 27 | 18 | PY01 | GPIO_Y_01(ALT0)<br>GPTMR3_CAPT_0(ALT1)<br>UART0_RXD(ALT2)<br>MCAN2_RXD(ALT7)<br>PWM0_P_1(ALT16)<br>PWM1_P_5(ALT17)<br>PWM0_FAULT_1(ALT18)<br>EWDG0_RST(ALT24)<br>USB0_OC(ALT25) | - | VPMC |
| 39 | 32 | 26 | - | PY02 | GPIO_Y_02(ALT0)<br>GPTMR3_COMP_1(ALT1)<br>UART0_DE(ALT2)<br>UART0_RTS(ALT3)<br>I2C2_SCL(ALT4)<br>MCAN2_STBY(ALT7)<br>PWM0_P_2(ALT16)<br>PWM1_P_6(ALT17)<br>ACMP_COMP_0(ALT18)<br>PWM1_FAULT_0(ALT19)<br>EWDG1_RST(ALT24)<br>USB0_PWR(ALT25) | - | VPMC |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 38 | 31 | 25 | - | PY03 | GPIO_Y_03(ALT0)<br>GPTMR3_CAPT_1(ALT1)<br>UART0_CTS(ALT3)<br>I2C2_SDA(ALT4)<br>MCAN3_STBY(ALT7)<br>PWM0_P_3(ALT16)<br>PWM1_P_7(ALT17)<br>ACMP_COMP_1(ALT18)<br>PWM1_FAULT_1(ALT19) | - | VPMC |
| 37 | 30 | - | - | PY04 | GPIO_Y_04(ALT0)<br>UART1_CTS(ALT3)<br>SPI2_CS_0(ALT5)<br>MCAN3_RXD(ALT7)<br>PWM0_P_4(ALT16)<br>TRGM0_P_04(ALT18) | - | VPMC |
| 36 | 29 | - | - | PY05 | GPIO_Y_05(ALT0)<br>GPTMR3_COMP_2(ALT1)<br>UART1_DE(ALT2)<br>UART1_RTS(ALT3)<br>SPI2_SCLK(ALT5)<br>MCAN3_TXD(ALT7)<br>PWM0_P_5(ALT16)<br>TRGM0_P_05(ALT18)<br>EWDG0_RST(ALT24) | - | VPMC |
| 35 | - | - | - | PY06 | GPIO_Y_06(ALT0)<br>GPTMR2_CAPT_0(ALT1)<br>UART1_RXD(ALT2)<br>I2C3_SDA(ALT4)<br>SPI2_MISO(ALT5)<br>PWM0_P_6(ALT16)<br>TRGM0_P_06(ALT18)<br>EWDG1_RST(ALT24) | - | VPMC |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 34 | - | - | - | PY07 | GPIO_Y_07(ALT0)<br>GPTMR2_COMP_0(ALT1)<br>UART1_TXD(ALT2)<br>I2C3_SCL(ALT4)<br>SPI2_MOSI(ALT5)<br>PWM0_P_7(ALT16)<br>TRGM0_P_07(ALT18) | - | VPMC |
| 7,17,20,44,57,69,83,94 | 5,15,36,47,54,73 | 5,12,39,45,58 | 5,9,28,33,45 | VDD_SOC | - | - | - |
| 8,10,16,22,32,43,56,66,70,76,84,95 | 6,12,17,27,35,46,55,68,74 | 6,13,23,38,46,59 | - | VSS | - | - | - |
| 9,15,85,96 | 11,67 | 7,47,60 | 6,34 | VIO_B00 | - | - | - |
| 21 | 16 | - | - | DCDC_SNS | - | - | - |
| 23 | 18 | 14 | 10 | VDD_OTPCAP | - | - | - |
| 26 | 21 | 17 | 13 | RESETN | - | - | - |
| 27 | 22 | 18 | 14 | WAKEUP | - | - | - |
| 28 | 23 | 19 | - | DCDC_GND | - | - | - |
| 29 | 24 | 20 | 15 | DCDC_LP | - | - | - |
| 30 | 25 | 21 | - | DCDC_IN | - | - | - |

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| Package | | | | PIN NAME | Digital Function | Analog Functi | Power Group |
|---|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | | |
| 31 | 26 | 22 | 16 | VPMC | - | - | - |
| 33 | 28 | 24 | 17 | VDD_PMCCAP | - | - | - |
| 42,55 | 45 | 37 | 29 | VIO_B01 | - | - | - |
| 64 | 50 | 42 | 30 | VREFL | - | - | - |
| 65 | 51 | 43 | 31 | VREFH | - | - | - |
| 67 | 52 | 44 | 32 | VANA | - | - | - |
| 68 | 53 | - | - | USBVBUS | - | - | - |
| 71 | 56 | - | - | VPLL | - | - | - |
| 72 | 57 | - | - | USB_DP | - | - | - |
| 73 | 58 | - | - | USB_DM | - | - | - |
| 74 | 59 | 48 | 35 | XTALI | - | - | - |
| 75 | 60 | 49 | 36 | XTALO | - | - | - |

Table 2: SOC IOMUX

| Package | | | | PIN NAME | Digital Function | Power Group |
|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | |
| 41 | 34 | 28 | 19 | PY00 | PGPIO_Y_00(ALT0)<br>PUART_TXD(ALT1)<br>PTMR_COMP_0(ALT2)<br>SOC_GPIO_Y_00(ALT3) | VPMC |
| 40 | 33 | 27 | 18 | PY01 | PGPIO_Y_01(ALT0)<br>PUART_RXD(ALT1)<br>PTMR_COMP_1(ALT2)<br>SOC_GPIO_Y_01(ALT3) | VPMC |
| 39 | 32 | 26 | - | PY02 | PGPIO_Y_02(ALT0)<br>PUART_RTS(ALT1)<br>PTMR_COMP_2(ALT2)<br>SOC_GPIO_Y_02(ALT3) | VPMC |

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| Package | | | | PIN NAME | Digital Function | Power Group |
|---|---|---|---|---|---|---|
| LQFP_100 | LQFP_80 | LQFP_64 | QFN_48 | | | |
| 38 | 31 | 25 | - | PY03 | PGPIO_Y_03(ALT0)<br>PUART_CTS(ALT1)<br>PTMR_COMP_3(ALT2)<br>SOC_GPIO_Y_03(ALT3) | VPMC |
| 37 | 30 | - | - | PY04 | PGPIO_Y_04(ALT0)<br>PTMR_COMP_0(ALT2)<br>SOC_GPIO_Y_04(ALT3) | VPMC |
| 36 | 29 | - | - | PY05 | PGPIO_Y_05(ALT0)<br>PEWDG_RST(ALT1)<br>PTMR_CAPT_0(ALT2)<br>SOC_GPIO_Y_05(ALT3) | VPMC |
| 35 | - | - | - | PY06 | PGPIO_Y_06(ALT0)<br>PTMR_COMP_1(ALT2)<br>SOC_GPIO_Y_06(ALT3) | VPMC |
| 34 | - | - | - | PY07 | PGPIO_Y_07(ALT0)<br>PTMR_CAPT_1(ALT2)<br>SOC_GPIO_Y_07(ALT3) | VPMC |

Table 3: PMIC IOMUX

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## 2.5 Special Function Pins

By default, the chip selects three different boot modes via the BOOT_MODE[1:0]=[PA03:PA02] pins. The boot configuration is shown in Table 4, and other special pin configurations are shown in Table 5.

| Boot Mode Selection Pins | | Boot Mode | Description |
|---|---|---|---|
| BOOT_MODE1 | BOOT_MODE0 | | |
| 0 | 0 | XPI NOR boot | boot from serial NOR FLASH connected to XPI0 |
| 0 | 1 | In-System Programming (ISP)/Serial Boot | Burn firmware from UART0/USB0, OTP, or boot from UART0/USB0 |
| 1 | 0 | In-System Programming (ISP)/Serial Boot | Burn firmware from UART0/USB0, OTP, or boot from UART0/USB0 |
| 1 | 1 | Retention Mode | Retention Mode |

Table 4: Startup configuration table

| Pin Name | Description | Recommended Usage |
|---|---|---|
| XTALI | 24MHz clock input | Connect to 24MHz crystal or active clock |
| XTALO | 24MHz clock output | Connect to 24MHz crystal or leave floating |

Table 5: Special Function Pin Configuration

_auto_part=0 Note: For LQFP64 and QFN48 packages of this product:

- USB_DP is multiplexed with PA24. When used for USB function, PA24 shall be configured as analog function (set the ANALOG bit of PAD[FUNC_CTL] to 1)
- USB_DM is multiplexed with PA25. When used for USB function, PA25 shall be configured as analog function (set the ANALOG bit of PAD[FUNC_CTL] to 1)
- When PA24 and PA25 are not used for USB, configure the USB PHY_CTRL0 register (PHY_CTRL0 |= 0x001000E0u) to disable the pull-down resistors of DP and DM.
- In In-System Programming (ISP) mode, only burning firmware from UART0 or booting from UART0 is supported

## 2.6 IO Reset Status

Table 6 summarizes the status of all IOs of this product after system reset:

| Name | Status after Reset | Function after Reset |
|---|---|---|
| PA04 | High-impedance | JTAG.TDO |

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| Name | Status after Reset | Function after Reset |
|---|---|---|
| PA05 | Input with internal pull-up | JTAG.TDI |
| PA06 | Input with internal pull-down | JTAG.TCK |
| PA07 | Input with internal pull-up | JTAG.TMS |
| PA08 | Input with internal pull-up | JTAG.TRST |
| PY00 | Output high level | PUART.TXD |
| PY01 | Input with internal pull-up | PUART.RXD |
| PY02 | Output high level | PUART.RTS |
| PY03 | Input with internal pull-up | PUART.CTS |
| PY05 | Output high level | PWDG.RST |
| Other IOs | Input state holder | GPIO |

Table 6: IO Reset Status Table

## 2.7 IO Power Supply Pins

For this product, the power supply pin for IOs of GPIO Port A (PA00~PA31) is VIO_B00.

The power supply pin for IOs of GPIO Port B (PB00~PB15) is VIO_B01.

The power supply pin for IOs of GPIO Port Y (PY00~PY07) is VPMC.

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# 3 Power Supply

This series of chips are powered by a single 3.0-3.6V power supply input to the DCDC_IN and VPMC pins, and provide the VDD_SOC, VDD_PMCCAP, VDD_OTPCAP required by the system through the built-in voltage regulator.

The VPMC pin supplies power for the power management domain and PDGO. Even if DCDC_IN is powered down, the contents of the power management domain and PDGO will still be retained.

Except for VIO_B00, each I/O power supply VIO_Bxx is connected to a 1.8V or 3.3V power supply according to the corresponding load. VIO_B00 is fixedly connected to a 3.3V power supply.

## 3.1 Power Supply Block Diagram

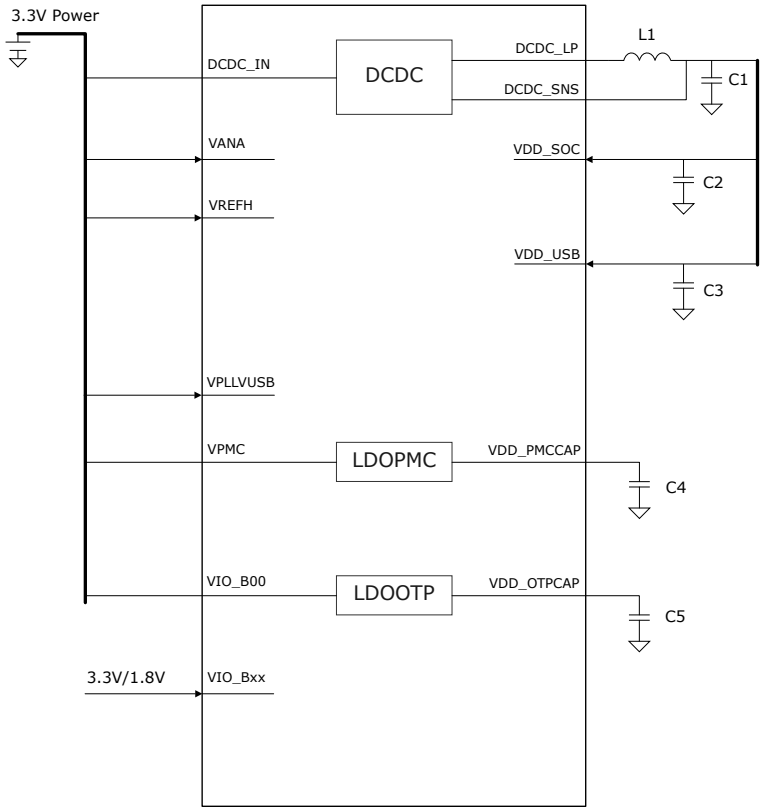



Figure 5: System Power Supply Block Diagram

The recommended values of inductors and capacitors are shown in Table 7.

| Designator | Reference Value |
|---|---|
| L1 | 2.2uH~10uF, typical value 4.7uH |
| C1 | 33~66uF |
| C2 | 0.1uF |
| C3 | 0.1uF |
| C4 | 4.7uF+0.1uF |
| C5 | 4.7uF+0.1uF |

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| Designator | Reference Value |
|---|---|

Table 7: Reference Values of Inductors and Capacitors for Power Supply Section

When the internal DCDC is bypassed and an external power supply is used for power supply, refer to the following block diagram. The recommended capacitor values are shown in Table 7.

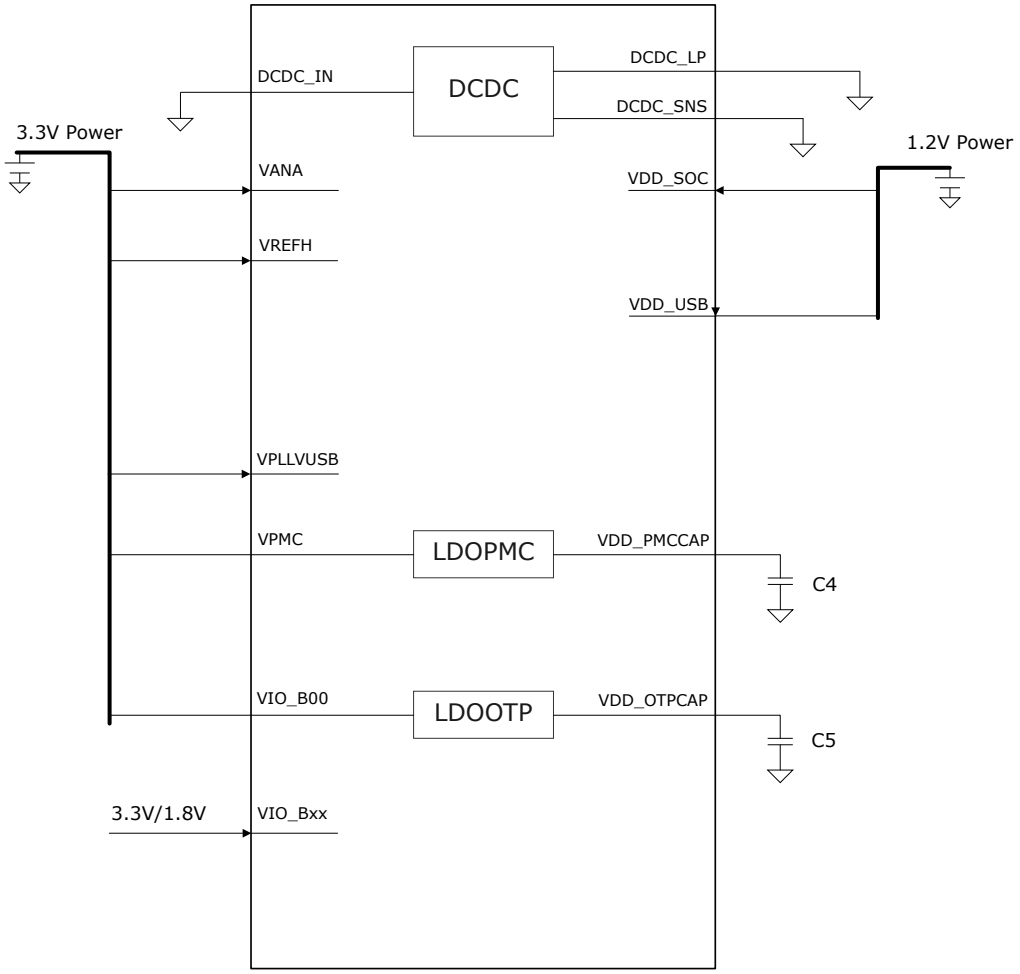



Figure 6: System Power Supply Block Diagram Without Built-in DCDC

Note: When using the built-in DCDC, the accuracy of the 16-bit ADC will decrease to a certain extent. For precautions for on-chip ADC usage, please refer to the ADC-related application documents at: https://www.hpmicro.com/design-resources/chip-information.

## 3.2 Power-up and Power-down Sequence

For power-up, it is only required that VPMC is not powered on later than other power supplies; for power-down, it is only required that VPMC is not powered off earlier than other power supplies.

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# 4 Electrical Characteristics

## 4.1 Operating Conditions

Unless otherwise specified, all voltages are referenced to VSS.

### 4.1.1 Absolute Maximum and Minimum Ratings

Table 8 lists the absolute maximum and minimum ratings of the operating environment supported by this chip. Exceeding the values listed in Table 8 may cause permanent damage to the chip.

| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| =0 DCDC_IN | DCDC input voltage | -0.3 | 3.6 | V |
| VPMC | VPMC input voltage | -0.3 | 3.6 | V |
| VDD_SOC | VDD_SOC input voltage | -0.3 | 1.3 | V |
| =1 VDD_USB | USB Core input voltage | -0.3 | 1.3 | V |
| VANA | VANA input voltage | -0.3 | 3.6 | V |
| VREFH | ADC reference voltage | 2.4 | 3.6 | V |
| =1 USB0_VBUS | USB0 input detection voltage | - | 5.5 | V |
| =1 VUSB | USB input voltage | -0.3 | 3.6 | V |
| =1 VUSBVPLL | USB and PLL input voltage | -0.3 | 3.6 | V |
| VIO_Bxx(3.3V mode) | 3.3V power supply for corresponding IOs | -0.3 | 3.6 | V |
| VIO_Bxx(1.8V mode) | 1.8V power supply for corresponding IOs | -0.3 | 1.98 | V |
| ESD HBM | ESD withstand voltage (HBM model) | - | 2000 | V |
| ESD CDM | ESD withstand voltage (CDM model) | - | 500 | V |
| T_S_T_G | Storage temperature | -40 | 150 | °C |

Table 8: Absolute Maximum and Minimum Ratings

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### 4.1.2 Recommended Operating Conditions

Table 9 lists the recommended operating conditions for the chip. Operating beyond the conditions listed in this table, the normal functionality and performance of the chip are not guaranteed.

| Symbol | Description | Operating Scenario | Max. Main Frequency | Min. | Typ. | Max.(1) | Unit |
|---|---|---|---|---|---|---|---|
| VDD_SOC(2) | VDD_SOC input voltage | Performance Mode | 480 MHz | 1.25 | 1.275 | 1.30 | V |
| | | Balanced Mode | 400 MHz | 1.15 | 1.175 | 1.30 | V |
| | | Low Power Mode | 280 MHz | 1.05 | 1.075 | 1.30 | V |
| =0 DCDC_IN | DCDC input voltage | - | - | 3.0 | 3.3 | 3.6 | V |
| VPMC | VPMC input voltage | - | - | 3.0 | 3.3 | 3.6 | V |
| VANA | VANA input voltage | - | - | 3.0 | 3.3 | 3.6 | V |
| =1 VBUS0 | VBUS0 input voltage | - | - | - | 5.0 | 5.5 | V |
| =1 VUSB | VUSB input voltage | - | - | 3.0 | 3.3 | 3.6 | V |
| =1 VUSBVPLL | USB and PLL input voltage | - | - | 3.0 | 3.3 | 3.6 | V |
| VIO_Bxx (3.3V mode) | 3.3V power supply for corresponding IOs | - | - | 3.0 | 3.3 | 3.6 | V |
| VIO_Bxx (1.8V mode) | 1.8V power supply for corresponding IOs | - | - | 1.62 | 1.8 | 1.98 | V |
| T_A | Ambient operating temperature | - | - | -40 | - | 105 | °C |
| T_J | Chip junction operating temperature | - | - | -40 | - | 125 | °C |

Table 9: Recommended Operating Conditions

1. Operating the chip at the maximum voltage will lead to high power consumption and heat generation. Long-term operation at maximum voltage will shorten the service life of the chip.

2. HPMICRO recommends setting the VDD_SOC voltage to the typical value under normal circumstances.

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## 4.2 Built-in Flash Characteristics

This product has a built-in 1MB flash memory.

The built-in 1MB flash memory is divided into 4096 pages, each containing 256 Bytes. A single programming operation can write up to 256 Bytes. Flash erasure can be performed per 4 pages (1K Byte), 16 pages (4K Bytes), 128 pages (32K Bytes), 256 pages (64K Bytes), or the entire chip. The characteristics of the built-in flash memory are shown in Table 10.

| Symbol | Description | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| ICC Standby | Flash standby current | - | 7.0 | 35 | uA |
| ICC DPD | Flash deep power down current | - | 0.6 | 4 | uA |
| ICC Read | Flash read current | - | 3 | 13 | mA |
| ICC P | Flash programming current | - | 1.5 | 4 | mA |
| ICC 1KE | Flash 1KB erasure current | - | 1 | 5 | mA |
| ICC 4KE | Flash 4KB erasure current | - | 1 | 6 | mA |
| ICC 32KE | Flash 32KB erasure current | - | 1 | 7 | mA |
| ICC 64KE | Flash 64KB erasure current | - | 1 | 8 | mA |
| ICC CE | Flash full-chip erasure current | - | 1 | 9 | mA |
| t BP | Flash byte programming time | - | 100 | 200 | us |
| t PP | Flash page programming time | - | 1 | 2 | ms |
| t SE | Flash 1KB/4KB sector erasure time | - | 2.3 | 5 | ms |
| t BE | Flash 32KB/64KB block erasure time | - | 2.3 | 5 | s |
| t CE | Flash full-chip erasure time | - | 5 | 9 | ms |
| Endurance | Program/erase cycles at 25°C | 200k | | - | Cycles |
| Data Retention | Data retention time at 25°C | - | 50 | - | Years |

Table 10: Built-in Flash Characteristics

## 4.3 DCDC Electrical Characteristics

The electrical characteristics of the built-in DCDC are shown in Table 11.

| Parameter | Minimum | Typical | Maximum | Unit | Remark |
|---|---|---|---|---|---|
| Input voltage | 3.0 | 3.3 | 3.6 | V | |
| Output voltage | 0.6 | - | 1.375 | V | |

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| Parameter | Minimum | Typical | Maximum | Unit | Remark |
|---|---|---|---|---|---|
| Output voltage accuracy (Run mode) | -3% | - | +3% | - | |
| Output voltage accuracy (LP mode) | -6% | - | +6% | - | |
| Overcurrent protection threshold | - | 2 | - | A | |
| Overvoltage protection threshold | - | 1.6 | - | V | |

Table 11: DCDC Electrical Characteristics

## 4.4 VPMC Undervoltage Detection

The characteristics of VPMC undervoltage detection BOR are shown in Table 12.

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Undervoltage Warning Assertion Voltage | VBOR Warning Assert | - | 2.8 | - | V | - |
| Undervoltage Warning Release Voltage | VBOR Warning Release | - | 2.9 | - | V | - |
| Undervoltage Reset Assertion Voltage | VBOR Reset Assert | - | 2.6 | - | V | - |
| Undervoltage Reset Release Voltage | VBOR Reset Release | - | 2.7 | - | V | - |

Table 12: Characteristics of VPMC Undervoltage Detection

## 4.5 Reset Pin RESET_N

For the requirement of the duration that RESET_N remains low to trigger a normal reset, please refer to Table 13.

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| RESET_N Low Level Duration | T resetn low | 300 | 1000 | - | us | - |

Table 13: Characteristics of RESET_N Low Level Reset

## 4.6 Oscillator

### 4.6.1 24MHz Oscillator Characteristics

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| Parameter | Symbol | Minimum | Typical | Maximum | Unit | Note |
|---|---|---|---|---|---|---|
| Frequency | FREQ | - | 24 | - | MHz | - |
| Equivalent Series Resistance | ESR | - | 40~80 | - | Ω | - |
| Load Capacitance | CL | - | 6 | - | pF | - |

Table 14: 24MHz Crystal Oscillator

### 4.6.2 32KHz RC Oscillator Clock Characteristics

| Parameter | Symbol | Minimum | Typical | Maximum | Unit | Note |
|---|---|---|---|---|---|---|
| Frequency | FREQ | - | 32 | - | KHz | - |
| Frequency Accuracy (Uncalibrated) | | -10 | - | 10 | % | - |

Table 15: 32KHz RC Oscillator

### 4.6.3 24MHz RC Oscillator Clock Characteristics

| Parameter | Symbol | Minimum | Typical | Maximum | Unit | Note |
|---|---|---|---|---|---|---|
| Frequency | FREQ | - | 24 | - | MHz | - |
| Frequency Accuracy | | -2 | - | 2 | % | - |

Table 16: 24MHz RC Oscillator

### 4.6.4 PLL Characteristics

| Parameter | Symbol | Minimum | Typical | Maximum | Unit | Note |
|---|---|---|---|---|---|---|
| Reference Frequency | fREF | - | 24 | - | MHz | - |
| VCO Frequency | fVCO | 400 | - | 1000 | MHz | - |
| Lock Time | tLOCK | - | - | 2400 | cycle | reference clock cycle |

Table 17: PLL Characteristic Parameters

## 4.7 Peripheral Clock Characteristics

Table 18 lists the clock characteristics of each peripheral of this product.

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| Symbol | Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| clk_top_cpu0 | 1.05V≤VDD_SOC≤1.30V | - | - | 480 | MHz |
| clk_top_mct0 | 1.05V≤VDD_SOC≤1.30V | - | - | 24 | MHz |
| clk_top_can0 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_can1 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_can2 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_can3 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_tmr0 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_tmr1 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_tmr2 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_tmr3 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_i2c0 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_i2c1 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_i2c2 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_i2c3 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_spi0 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_spi1 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_spi2 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_spi3 | 1.05V≤VDD_SOC≤1.30V | - | - | 80 | MHz |
| clk_top_urt0 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt1 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt2 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt3 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt4 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt5 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt6 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_urt7 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_xpi0 | 1.05V≤VDD_SOC≤1.30V | - | - | 333 | MHz |
| clk_top_ana0 | 1.05V≤VDD_SOC≤1.30V | - | - | 200 | MHz |
| clk_top_ana1 | 1.05V≤VDD_SOC≤1.30V | - | - | 200 | MHz |
| clk_top_ana2 | 1.05V≤VDD_SOC≤1.30V | - | - | 200 | MHz |
| clk_top_ana3 | 1.05V≤VDD_SOC≤1.30V | - | - | 200 | MHz |
| clk_top_ref0 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |
| clk_top_ref1 | 1.05V≤VDD_SOC≤1.30V | - | - | 100 | MHz |

Table 18: Peripheral Clock Characteristics

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## 4.8 Operating Modes

The power configuration of each module of the chip under different modes is shown in Table 19

| Mode | CPU0 Subsystem Power | VDD_SOC | VPMC | DGO |
|---|---|---|---|---|
| Wait Mode | On | On | On | On |
| Stop Mode | Optional | On | On | On |
| Sleep Mode | Off | Off | On | On |
| Shutdown Mode | Off | Off | Off | On |

Table 19: Operating Mode Configuration Table

## 4.9 Power Supply Current Characteristics

Current consumption is affected by multiple parameters and factors, including operating voltage, ambient temperature, I/O pin load, device software configuration, operating frequency, I/O pin switching rate, program location in memory, running code, etc.

The supply current of IDD(DCDC_IN) is shown in Table 20. DCDC_IN and VPMC are powered by an external 3.3V supply, and VDD_SOC is generated by the on-chip DCDC. The CPU runs the CoreMark program, and the code is executed from the Instruction Local Memory (ILM). All peripherals operate at the default frequency after their clocks are enabled (please refer to the HPM5300 User Manual for details). All tests are performed on chips with typical process parameters, for reference only.

The supply current of IDD(VPMC) is shown in Table 21.

The supply current of IDD(VANA) is shown in Table 22.

The supply current of IDD(VPLL) is shown in Table 23.

| Symbol | Test Condition | CPU0 | Peripheral Status | $T_A$=25°C | $T_A$=85°C | $T_A$=105°C | Unit |
|---|---|---|---|---|---|---|---|
| IDD DCDC_IN = 3.3V | VDD_SOC=1.175V CPU/BUS@480/160MHz | On | All Enabled | 59.9 | 64.7 | 67.8 | mA |
| | | On | All Disabled | 33.5 | 36.8 | 39.6 | mA |
| | VDD_SOC=1.10V CPU/BUS@240/120MHz | On | All Enabled | 36.2 | 39.1 | 41.4 | mA |
| | | On | All Disabled | 18.3 | 20.7 | 22.8 | mA |
| | VDD_SOC=1.10V CPU/BUS@100/100MHz | On | All Enabled | 28.7 | 31.3 | 33.6 | mA |
| | | On | All Disabled | 13.1 | 15.4 | 17.3 | mA |

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| Symbol | Test Condition | CPU0 | Peripheral Status | $T_A$=25°C | $T_A$=85°C | $T_A$=105°C | Unit |
|---|---|---|---|---|---|---|---|
| | VDD_SOC=1.05V CPU/BUS@24/24MHz | On | All Enabled | 5.5 | 7.3 | 8.8 | mA |
| | | On | All Disabled | 3.2 | 4.8 | 6.4 | mA |

Table 20: Typical Current in Operating Modes

| Symbol | Test Condition | Operating Status | $T_A$=25°C | $T_A$=85°C | $T_A$=105°C | Unit |
|---|---|---|---|---|---|---|
| IDD PMC | VPMC = 3.3V | Operating Mode DCDC ON | 1.06 | 1.14 | 1.20 | mA |
| IDD PMC | VPMC = 3.3V | Sleep Mode DCDC OFF | 0.53 | 0.59 | 0.64 | mA |
| IDD PMC | VPMC = 3.3V | Shutdown Mode DGO Only | 2.3 | 3.1 | 4.8 | uA |

Table 21: Typical IDD(VPMC) Current

| Symbol | Test Condition | Operating Status | $T_A$=25°C | $T_A$=85°C | $T_A$=105°C | Unit |
|---|---|---|---|---|---|---|
| IDD VANA | VANA = 3.3V | ADC0 10KSPS | 0.17 | 0.18 | 0.19 | mA |
| IDD VANA | VANA = 3.3V | All Analog Modules Disabled | 0.09 | 0.10 | 0.10 | mA |

Table 22: Typical IDD(VANA) Current

| Symbol | Test Condition | Operating Status | $T_A$=25°C | $T_A$=85°C | $T_A$=105°C | Unit |
|---|---|---|---|---|---|---|
| IDD VPLL | VPLL = 3.3V | All PLLs Enabled | 2.55 | 2.64 | 2.67 | mA |
| IDD VPLL | VPLL = 3.3V | PLL0 Enabled | 1.75 | 1.79 | 1.81 | mA |
| IDD VPLL | VPPL = 3.3V | All PLLs Disabled | 0.83 | 0.83 | 0.84 | mA |

Table 23: Typical IDD(VPLL) Current

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## 4.10 I/O Characteristics

### 4.10.1 I/O DC Characteristics

I/O characteristics are shown in Table 24.

| Symbol | Parameter | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| VDDIO 1.8V | IO supply voltage | 1.62 | 1.8 | 1.98 | V |
| VDDIO 3.3V | IO supply voltage | 2.97 | 3.3 | 3.63 | V |
| VIL 1.8V | Input low level voltage | 0 | - | 0.3*VDDIO | V |
| VIH 1.8V | Input high level voltage | 0.7*VDDIO | - | VDDIO | V |
| VOL 1.8V | Output low level voltage | - | - | 0.15 | V |
| VOH 1.8V | Output high level voltage | VDDIO-0.15 | - | - | V |
| VIL 3.3V | Input low level voltage | 0 | - | 0.3*VDDIO | V |
| VIH 3.3V | Input high level voltage | 0.7*VDDIO | - | VDDIO | V |
| VOL 3.3V | Output low level voltage | - | - | 0.15 | V |
| VOH 3.3V | Output high level voltage | VDDIO-0.15 | - | - | V |
| RPU22K | Pull-up resistor | 17.1 | 22 | 28.3 | kΩ |
| RPU47K | Pull-up resistor | 36 | 47 | 60 | kΩ |
| RPU100K | Pull-up resistor | 75 | 100 | 125 | kΩ |
| RPD100K | Pull-down resistor | 75 | 100 | 125 | kΩ |

Table 24: IO Operating Conditions

### 4.10.2 I/O AC Characteristics

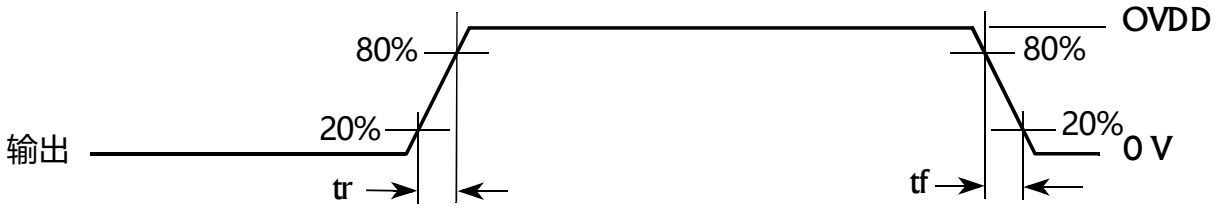



Figure 7: I/O AC Characteristics

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| Type | Parameter | Symbol | Min. | Max. | Unit | Test Conditions |
|---|---|---|---|---|---|---|
| 3.3/1.8V IO @1.8V | Rise/Fall time | tr/tf | - | 4.4/4.3 | ns | 15pF load, fast slew rate, drive strength 111b |
| 3.3/1.8V IO @1.8V | Rise/Fall time | tr/tf | - | 8.2/7.9 | ns | 15pF load, slow slew rate, drive strength 111b |
| 3.3/1.8V IO @1.8V | Rise/Fall time | tr/tf | - | 4.6/4.4 | ns | 15pF load, fast slew rate, drive strength 011b |
| 3.3/1.8V IO @1.8V | Rise/Fall time | tr/tf | - | 8.6/8.3 | ns | 15pF load, slow slew rate, drive strength 011b |
| 3.3/1.8V IO @3.3V | Rise/Fall time | tr/tf | - | 2.6/2.5 | ns | 15pF load, fast slew rate, drive strength 111b |
| 3.3/1.8V IO @3.3V | Rise/Fall time | tr/tf | - | 4.3/4.2 | ns | 15pF load, slow slew rate, drive strength 111b |
| 3.3/1.8V IO @3.3V | Rise/Fall time | tr/tf | - | 2.9/2.7 | ns | 15pF load, fast slew rate, drive strength 011b |
| 3.3/1.8V IO @3.3V | Rise/Fall time | tr/tf | - | 4.5/4.4 | ns | 15pF load, slow slew rate, drive strength 011b |
| 3.3V IO @3.3V | Rise/Fall time | tr/tf | - | 2.1/1.6 | ns | 15pF load, fast slew rate, drive strength 111b |
| 3.3V IO @3.3V | Rise/Fall time | tr/tf | - | 3.4/3.3 | ns | 15pF load, slow slew rate, drive strength 111b |
| 3.3V IO @3.3V | Rise/Fall time | tr/tf | - | 2.2/1.7 | ns | 15pF load, fast slew rate, drive strength 011b |
| 3.3V IO @3.3V | Rise/Fall time | tr/tf | - | 3.6/3.4 | ns | 15pF load, slow slew rate, drive strength 011b |

Table 25: I/O AC Characteristics

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## 4.11 JTAG Interface

The JTAG timing sequence is shown in Figure 8.

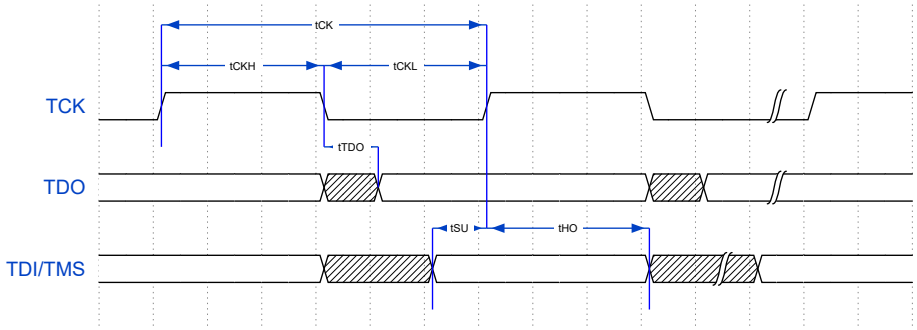



Figure 8: JTAG Timing Diagram

| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| tCK | Duration of one clock cycle | 40 | - | ns |
| tCKH | High level duration in one clock cycle | 0.48*P | 0.52*P | ns |
| tCKL | Low level duration in one clock cycle | 0.48*P | 0.52*P | ns |
| tSU(TDI-TCK) | Input setup time, from TCK high to TDI valid | 8 | - | ns |
| tSU(TMS-TCK) | Input setup time, from TCK high to TMS valid | 8 | - | ns |
| tHO(TCK-TDI) | Input hold time, from TCK high to TDI valid | 15 | - | ns |
| tHO(TCK-TMS) | Input hold time, from TCK high to TMS valid | 15 | - | ns |
| tTDO(TCK-TDO) | Time from TCK falling edge to TDO data valid | - | 15 | ns |

Table 26: JTAG Timing Parameters

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## 4.12 XPI Memory Interface

### 4.12.1 DC Characteristics

Refer to I/O characteristics.

### 4.12.2 AC Characteristics

There are three sources for the XPI sampling clock:

- Generated by the XPI controller and looped back via DQS (XPI_GCR0[RXCLKSRC] = 0x0)
- Generated by the XPI controller and looped back internally (XPI_GCR0[RXCLKSRC] = 0x1)
- Input from external DQS (XPI_GCR0[RXCLKSRC] = 0x3)

The following describes the characteristics and timing of input/read operations corresponding to the three sampling clock sources and SDR/DDR modes. The measurement data is based on a capacitive load of 15pF and an input slew rate of 1V/ns.

#### 4.12.2.1 SDR Mode

The timing for XPI_GCR0[RXCLKSRC] = 0x0, 0x1 is shown in Figure 9.

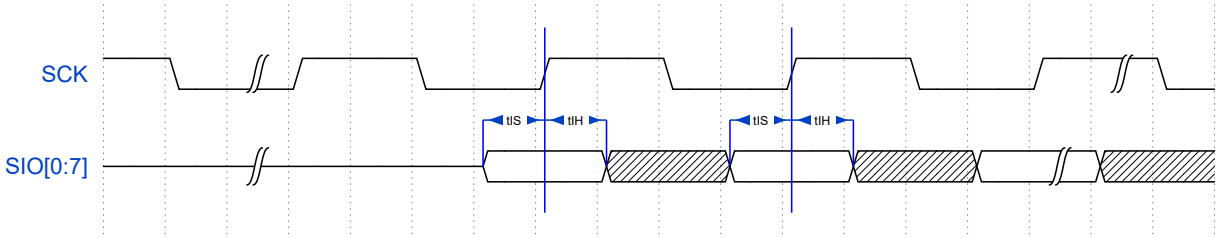



Figure 9: XPI SDR Mode Input Timing (XPI_GCR0[RXCLKSRC] = 0x0, 0x1)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 60 | MHz |
| tIS | Input data setup time | 8.67 | — | ns |
| tIH | Input data hold time | 0 | — | ns |

Table 27: XPI SDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x0)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 133 | MHz |
| tIS | Input data setup time | 2 | — | ns |
| tIH | Input data hold time | 1 | — | ns |

Table 28: XPI SDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x1)

The timing shown in Figure 9 is based on the memory generating read data on the falling edge of SCK, and the XPI controller sampling the read data on the falling edge.

In SDR mode, when XPI_GCR0[RXCLKSRC] = 0x3 and the memory provides read data and read strobe, there are two scenarios:

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- Scenario 1: The memory generates read data and read strobe signals on the rising (or falling) edge of SCK.

  The timing for XPI_GCR0[RXCLKSRC] = 0x3, Scenario 1 is shown in Figure 10.

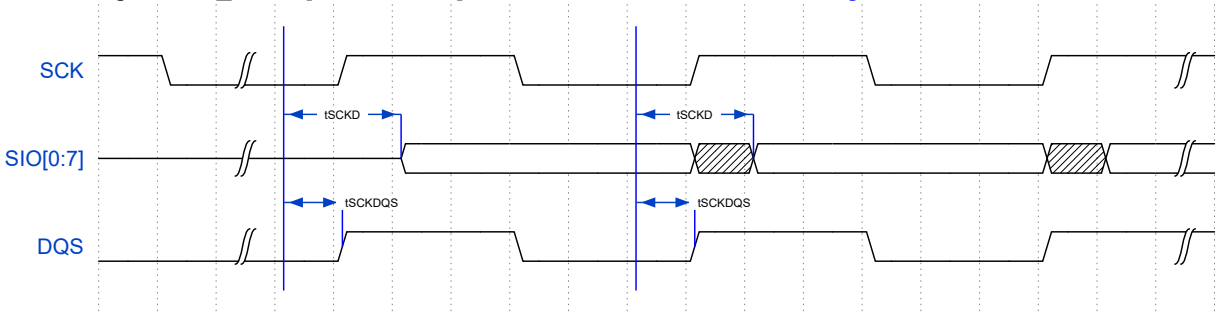



Figure 10: XPI SDR Mode Input Timing (XPI_GCR0[RXCLKSRC] = 0x3, Scenario 1)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 166 | MHz |
| tSCKD - tSCKDQS | Time difference between tSCKD and tSCKDQS | -2 | 2 | ns |

Table 29: XPI SDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x3, Scenario 1)

The timing shown in Figure 10 is based on the memory generating read data and read strobe on the rising edge of SCK, and the XPI controller sampling the read data on the falling edge of DQS.

- Scenario 2: The memory generates read data on the falling edge of SCK and read strobe on the rising edge of SCK.

  The timing for XPI_GCR0[RXCLKSRC] = 0x3, Scenario 2 is shown in Figure 11.

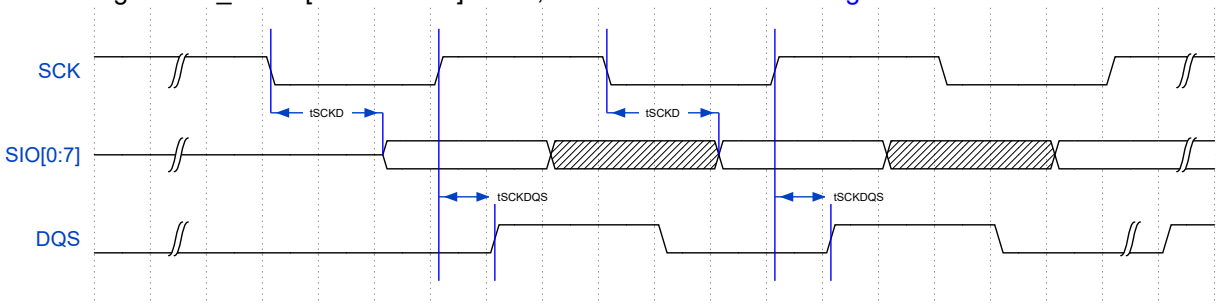



Figure 11: XPI SDR Mode Input Timing (XPI_GCR0[RXCLKSRC] = 0x3, Scenario 2)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 166 | MHz |
| tSCKD - tSCKDQS | Time difference between tSCKD and tSCKDQS | -2 | 2 | ns |

Table 30: XPI SDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x3, Scenario 2)

The timing shown in Figure 11 is based on the memory generating read data on the falling edge of SCK and read strobe on the rising edge of SCK, and the XPI controller sampling the read data on the rising edge of DQS with a half-cycle delay.

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#### 4.12.2.2 DDR Mode

The input timing of XPI DDR mode (XPI_GCR0[RXCLKSRC] = 0x0, 0x1) is shown in Figure 12.

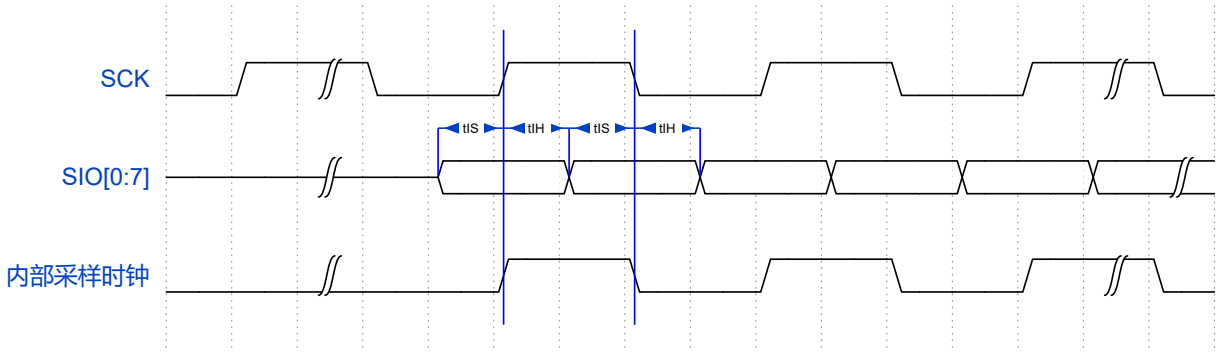



Figure 12: XPI DDR Mode Input Timing (XPI_GCR0[RXCLKSRC] = 0x0, 0x1)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 30 | MHz |
| tIS | Input data setup time | 8.67 | — | ns |
| tIH | Input data hold time | 0 | — | ns |

Table 31: XPI DDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x0)

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 66 | MHz |
| tIS | Input data setup time | 2 | — | ns |
| tIH | Input data hold time | 1 | — | ns |

Table 32: XPI DDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x1)

In DDR mode, when XPI_GCR0[RXCLKSRC] = 0x3, the memory generates read data and read strobe signals on the rising (or falling) edge of SCK.

The input timing of XPI DDR mode (XPI_GCR0[RXCLKSRC] = 0x3) is shown in Figure 13.

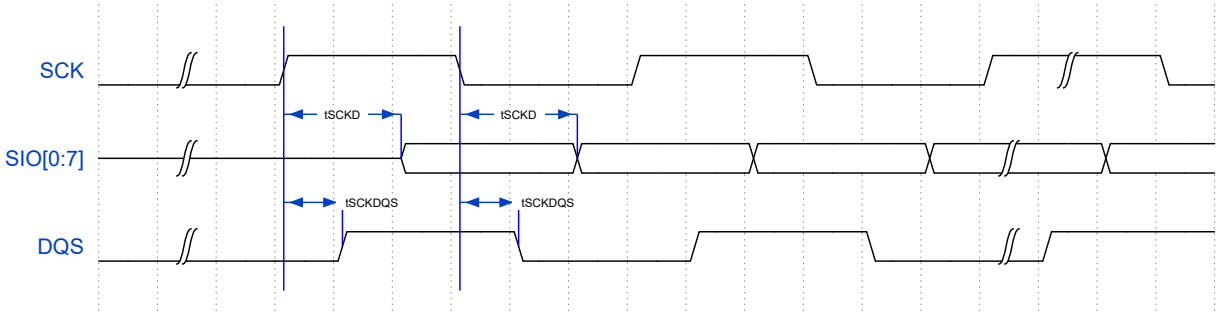



Figure 13: XPI DDR Mode Input Timing (XPI_GCR0[RXCLKSRC] = 0x3)

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| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| | Clock frequency | — | 166 | MHz |
| tSCKD - tSCKDQS | Time difference between tSCKD and tSCKDQS | -1 | 1 | ns |

Table 33: XPI DDR Mode Input Characteristics (XPI_GCR0[RXCLKSRC] = 0x3)

#### 4.12.2.3 XPI Output/Write Operation

The following section describes the output signal timing of the XPI controller, including control signals and data output.

- SDR Mode

  The output signal timing of XPI SDR mode is shown in Figure 14.

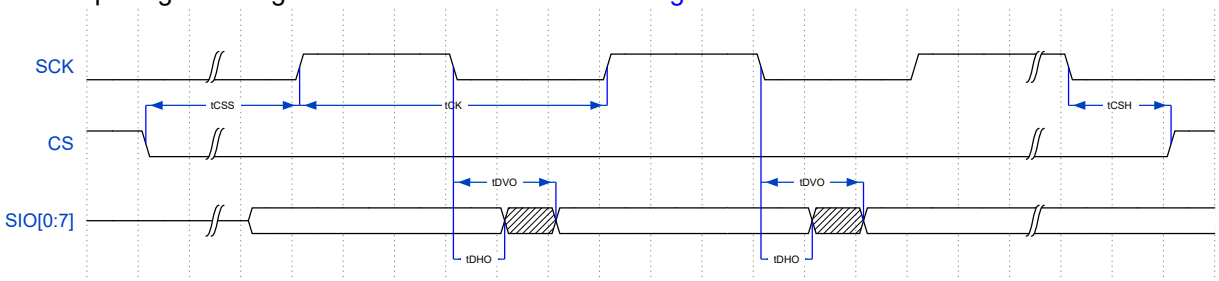



Figure 14: XPI SDR Mode Output Signals

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| fCK | Clock frequency | — | 166 | MHz |
| tCK | SCK clock period | 6 | — | ns |
| tDVO | Output signal valid time | — | 1 | ns |
| tDHO | Output signal hold time | 1 | — | ns |
| tCSS | Chip select setup time | 3 x tCK - 1 | — | ns |
| tCSH | Chip select hold time | 3 x tCK + 2 | — | ns |

Table 34: XPI SDR Mode Output Signal Timing

- DDR Mode

  The output signal timing of XPI DDR mode is shown in Figure 15.

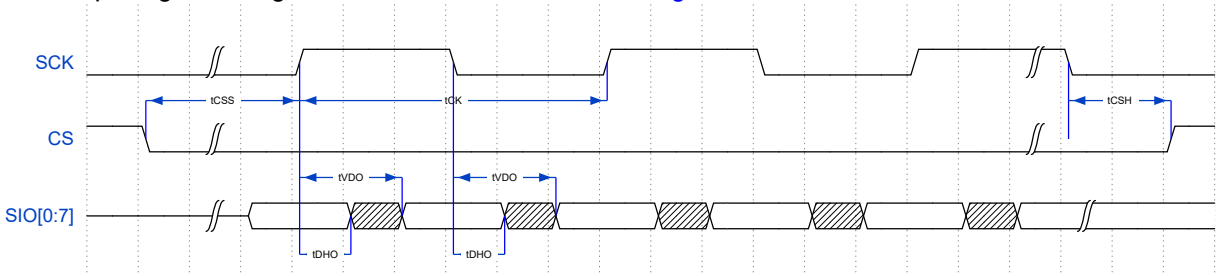



Figure 15: XPI DDR Mode Output Signals

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| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| fCK | Clock frequency | — | 166 | MHz |
| tCK | SCK clock period (XPI_GCR0[RXCLKSRC] = 0x0) | 6 | — | ns |
| tDVO | Output signal valid time | — | 2.2 | ns |
| tDHO | Output signal hold time | 0.8 | — | ns |
| tCSS | Chip select setup time | 3 x tCK/2 - 0.7 | — | ns |
| tCSH | Chip select hold time | 3 x tCK/2 + 0.8 | — | ns |

Table 35: XPI DDR Mode Output Signal Timing

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## 4.13 Analog Interface

### 4.13.1 16-bit Analog-to-Digital Converter (ADC) Characteristics

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Supply Voltage | VDDA | 3 | 3.3 | 3.6 | V | - |
| Input Signal Voltage | Vin | VREFL | - | VREFH | V | - |
| Input Sampling Capacitance | Cs | - | 4 | - | pF | - |
| Sampling Switch On-Resistance | Ron | - | 300 | - | ohm | - |
| Reference High Level | VREFH | 2.4 | - | VDDA | V | - |
| Reference Low Level | VREFL | 0 | - | - | V | - |
| Sampling Rate | fs | - | 2 | - | MHz | - |
| Differential Non-Linearity | DNL | - | +1/-0.89 | - | LSB | Single-ended signal |
| Integral Non-Linearity | INL | - | +3.1/-5.2 | - | LSB | Single-ended signal |
| Offset Error | Vos | - | 4 | - | LSB | Single-ended signal, input connected to ground |
| Gain Error (Full Swing Error) | GE | - | 3 | - | LSB | Single-ended signal, input connected to VREFH |
| Total Unadjusted Error | TUE | - | 7.2 | - | - | Single-ended signal |
| Signal to Noise and Distortion Ratio | SINAD | - | 74 | - | dB | Single-ended signal (VREFH=3.0V) |
| Effective Number of Bits | ENOB | - | 12 | - | Bit | Single-ended signal (VREFH=3.0V) |
| Total Harmonic Distortion | THD | - | 93 | - | dB | Single-ended signal |

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| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|

Table 36: 16-bit ADC Parameters

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### 4.13.2 Analog Comparator (ACMP) Characteristics

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Supply Voltage | VDDA | 3 | 3.3 | 3.6 | V | - |
| Input Signal Level | Vin | 0 | - | VDDA | V | - |
| Input Offset | Vos | -3 | - | 3 | mV | HPMODE=1 |
| | | -6 | - | 6 | mV | HPMODE=0 |
| Hysteresis Voltage | Vhyst | 18 | 24 | 30 | mV | HPMODE=0; HYST<1:0>=00 |
| | | 12 | 16 | 20 | mV | HPMODE=0; HYST<1:0>=01 |
| | | 6 | 8 | 10 | mV | HPMODE=0; HYST<1:0>=10 |
| | | 0 | 0 | 0 | mV | HPMODE=0; HYST<1:0>=11 |
| | | 24 | 30 | 36 | mV | HPMODE=1; HYST<1:0>=00 |
| | | 16 | 20 | 24 | mV | HPMODE=1;HYST<1:0>=01 |
| | | 8 | 10 | 12 | mV | HPMODE=1;HYST<1:0>=10 |
| | | 0 | 0 | 0 | mV | HPMODE=1;HYST<1:0>=11 |
| Propagation Delay | Tp | 60 | 80 | 100 | ns | HPMODE=0 |
| | | 5 | 6.5 | 8.5 | ns | HPMODE=1 |

Table 37: Comparator Parameters

### 4.13.3 12-bit Digital-to-Analog Converter (DAC) Characteristics

| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| VDDA | Supply Voltage | 3.0 | 3.3 | 3.6 | V |
| VDACR | Reference Voltage | 1.71 | 3.6 | 3.6 | V |
| CLD | Output Load Capacitance | - | - | 400 | pF |
| RLD | Output Load Resistance | 3k | - | - | ohm |
| IDDA DACLP | DAC Low Power Mode Supply Current, no load | - | 360 | - | uA |
| IDDA DACHP | DAC High Performance Mode Supply Current, no load | - | 1.6 | - | mA |
| Tpwrup | DAC Power-up Time to Valid Output | - | - | 10 | uA |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Tdachp | DAC High Performance Mode Output Settling Time<br>RLD = 3K ohm，CLD = 400 pF<br>Digital code 410~3891 | - | 1 | 1.2 | uS |
| Tdaclp | DAC Low Power Mode Output Settling Time<br>RLD = 3K ohm，CLD = 400 pF<br>Digital code 410~3891 | - | 3.5 | 4 | uS |
| SRHP | DAC High Performance Mode Slew Rate<br>RLD = 3K ohm，CLD = 400 pF<br>Digital code 410~3891 | - | 3 | - | V/us |
| SRLP | DAC Low Power Mode Slew Rate<br>RLD = 3K ohm，CLD = 400 pF<br>Digital code 410~3891 | - | 7.2 | - | V/us |
| INL | Integral Non-Linearity | - | ±3 | - | LSB |
| DNL | Differential Non-Linearity | - | ±0.8 | - | LSB |
| voffset | Offset Error<br>Digital code 410~3891 | - | ±6 | - | mV |
| Egain | Gain Error | - | ±0.5 | - | % |
| Vout | Output Voltage Range | VSSA+0.04 | - | VDDA-0.04 | V |

Table 38: 12-bit DAC Parameters

### 4.13.4 Operational Amplifier (OPAMP) Characteristics

| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| VDDA | Supply Voltage | 3.0 | 3.3 | 3.6 | V |
| CMIR | Common Mode Input Range | 0 | - | VDDA | V |
| Vloffset | Input Offset Voltage<br>Vcm = VDDA/2, all temperature，no load | - | - | ±2 | mV |
| Vloffset | Input Offset Voltage<br>All common mode input range, all temperature，no load | - | - | ±3 | mV |
| Iload | Drive Current，source or sink current<br>250mV≤output Voltage≤VDDA-250mV | - | - | 3300 | uA |
| Rload | Resistive Load | 1k | - | - | ohm |
| Cload | Capacitive Load | 1 | - | - | pF |
| CMRR | Common Mode Rejection Ratio | - | 76 | - | dB |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| PSRR | Power Supply Rejection Ratio<br>1KHz, Cload = 50pF,<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | 83 | 87 | - | dB |
| PSRR | Power Supply Rejection Ratio<br>1KHz, Cload = 50pF,<br>250mV≤output Voltage≤VDDA-250mV<br>Non-inv amplify Configuration | 55 | 77 | - | dB |
| PSRR | Power Supply Rejection Ratio<br>1KHz, Cload = 50pF,<br>250mV≤output Voltage≤VDDA-250mV<br>Inv amplify Configuration | 50 | 78 | - | dB |
| GBW | Gain bandwidth product<br>Cload≤50pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | 3 | 10 | 16 | MHz |
| GBW | Gain bandwidth product<br>Cload≤5pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | 7 | 27 | 43 | MHz |
| SR | Slew Rate<br>From 10% to 90% Output voltage,<br>Cload≤50pF<br>Follower Configuration | - | 7 | - | V/us |
| SR | Slew Rate<br>From 10% to 90% Output voltage,<br>Cload≤5pF<br>Follower Configuration | - | 18 | - | V/us |
| AO | Open Loop Gain<br>No Rload,<br>250mV≤output Voltage≤VDDA-250mV | 92 | 115 | 118 | dB |
| AO | Open Loop Gain<br>Rload≥1Kohm,<br>250mV≤output Voltage≤VDDA-250mV | 92 | 115 | 118 | dB |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| PM | Phase Margin<br>Cload≤50pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | - | 76 | - | ○ |
| PM | Phase Margin<br>Cload≤5pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | - | 66 | - | ○ |
| GM | Gain Margin<br>Cload≤50pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | - | 16 | - | dB |
| PM | Gain Margin<br>Cload≤5pF,<br>No Rload or Rload≥1Kohm<br>250mV≤output Voltage≤VDDA-250mV<br>Follower Configuration | - | 11 | - | dB |
| Vohsat | High Saturation Output Voltage | VDDA-150mV | - | - | mV |
| Volsat | Low Saturation Output Voltage | - | - | 150 | mV |
| Twakeup | Wake up time from disable state<br>Cload≤50pF, Rload≥1Kohm<br>Follower Configuration | - | 1.7 | 2.8 | us |
| Twakeup | Wake up time from disable state<br>Cload≤5pF, Rload≥1Kohm<br>Follower Configuration | - | 1.5 | 3.2 | us |
| Gerr | Gain Error<br>PGA Gain = 2,<br>250mV≤output Voltage≤VDDA-250mV | 0 | - | 0.5 | % |
| Gerr | Gain Error<br>PGA Gain = 4,<br>250mV≤output Voltage≤VDDA-250mV | -0.3 | - | 0.2 | % |
| Gerr | Gain Error<br>PGA Gain = 8,<br>250mV≤output Voltage≤VDDA-250mV | -0.9 | - | 0.1 | % |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Gerr | Gain Error<br>PGA Gain = 16,<br>250mV≤output Voltage≤VDDA-250mV | -2.2 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = 32,<br>250mV≤output Voltage≤VDDA-250mV | -5.3 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = 64,<br>250mV≤output Voltage≤VDDA-250mV | -7.2 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = 128,<br>250mV≤output Voltage≤VDDA-250mV | -10.3 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = -2,<br>250mV≤output Voltage≤VDDA-250mV | -0.9 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = -4,<br>250mV≤output Voltage≤VDDA-250mV | -0.9 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = -8,<br>250mV≤output Voltage≤VDDA-250mV | -1 | - | 0 | % |
| Gerr | Gain Error<br>PGA Gain = -16,<br>250mV≤output Voltage≤VDDA-250mV | -1.2 | - | 0.3 | % |
| Gerr | Gain Error<br>PGA Gain = -32,<br>250mV≤output Voltage≤VDDA-250mV | -1.7 | - | 0.8 | % |
| Gerr | Gain Error<br>PGA Gain = -64,<br>250mV≤output Voltage≤VDDA-250mV | -3 | - | 1.8 | % |
| Gerr | Gain Error<br>PGA Gain = -128,<br>250mV≤output Voltage≤VDDA-250mV | -5.8 | - | 3.7 | % |
| Rpga | PGA R2/R1 resistor value,<br>non-inverting configuration,<br>PGA gain = 2 | - | 10/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value,<br>non-inverting configuration,<br>PGA gain = 4 | - | 30/10 | - | Kohm/Kohm |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Rpga | PGA R2/R1 resistor value, non-inverting configuration, PGA gain = 8 | - | 70/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, non-inverting configuration, PGA gain = 16 | - | 150/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, non-inverting configuration, PGA gain = 32 | - | 310/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, non-inverting configuration, PGA gain = 64 | - | 630/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, non-inverting configuration, PGA gain = 128 | - | 1270/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -2 | - | 20/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -4 | - | 40/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -8 | - | 80/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -16 | - | 160/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -32 | - | 320/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -64 | - | 640/10 | - | Kohm/Kohm |
| Rpga | PGA R2/R1 resistor value, inverting configuration, PGA gain = -128 | - | 1280/10 | - | Kohm/Kohm |
| Delta R | Resisitor Variation | -5 | - | 5 | % |
| PGA BW | PGA Bandwith, non-inverting configuration, PGA gain = 2~128 | - | GBW/gain | - | MHz |

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| Symbol | Description/Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| PGA BW | PGA Bandwith,<br>inverting configuration,<br>PGA gain = -2~-128 | - | GBW/(\|$gain$\|+1) | - | MHz |
| Ndensity | Voltage Noise Density<br>1KHz, Rload = 1Kohm | - | 55 | - | nV/$\sqrt{Hz}$ |
| Ndensity | Voltage Noise Density<br>10KHz, Rload = 1Kohm | - | 22 | - | nV/$\sqrt{Hz}$ |
| IDD | OPAMP Operating Current<br>No Rload，Follower Configuration | - | 810 | - | uA |

Table 39: OPAMP Parameters

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## 4.14 SPI Interface

### 4.14.1 SPI Master Mode Timing Diagram

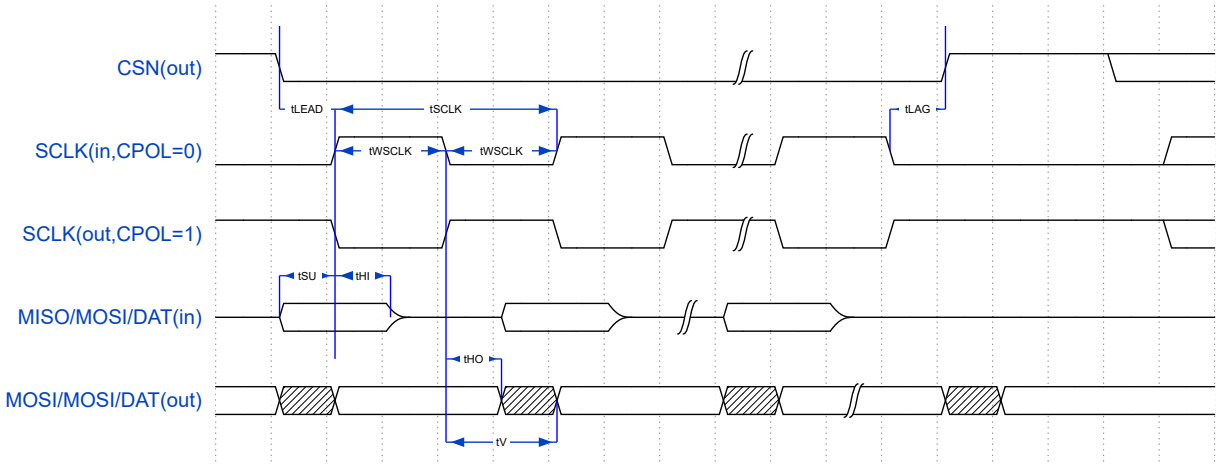



Figure 16: SPI Master Mode Timing (CPHA=0)

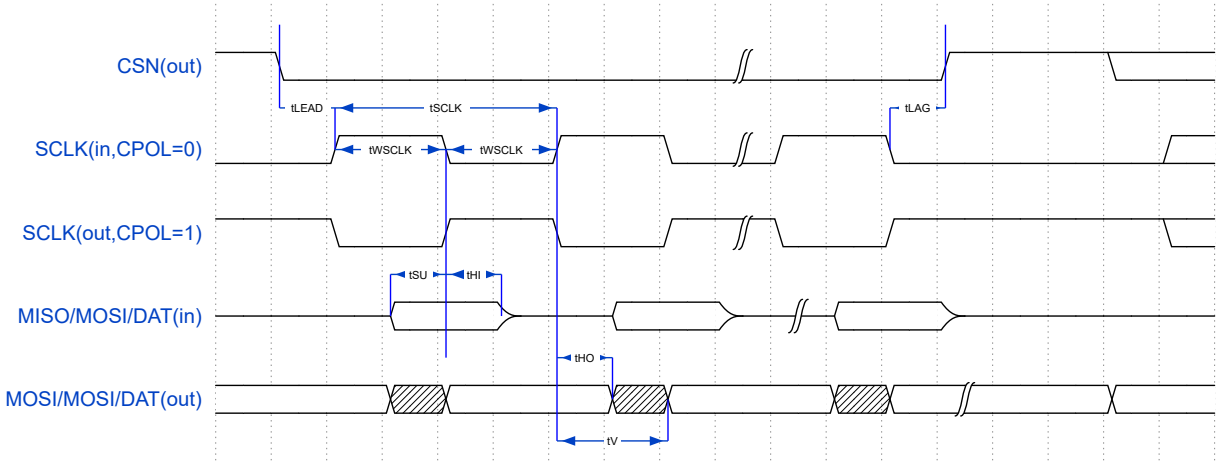



Figure 17: SPI Master Mode Timing (CPHA=1)

| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| tSCK | SCK Period | 12.5 | — | ns |
| tLEDA | CS Setup Time | 1 | — | tperiph |
| tLAG | CS Hold Time | 1 | — | tperiph |
| tWSCK | Clock High or Low Duration | tSCK / 2 - 3 | — | ns |
| tSU | Data Setup Time (Input) | 10 | — | ns |
| tHI | Data Hold Time (Input) | 2 | — | ns |

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| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| tV | Data Valid (Delay from SCLK Edge) | — | 8 | ns |
| tHO | Data Hold Time (Output) | 0 | — | ns |

Table 40: SPI Master Mode Parameters (Note: tperiph = 1000 / fperiph)

### 4.14.2 SPI Slave Mode Timing Diagram

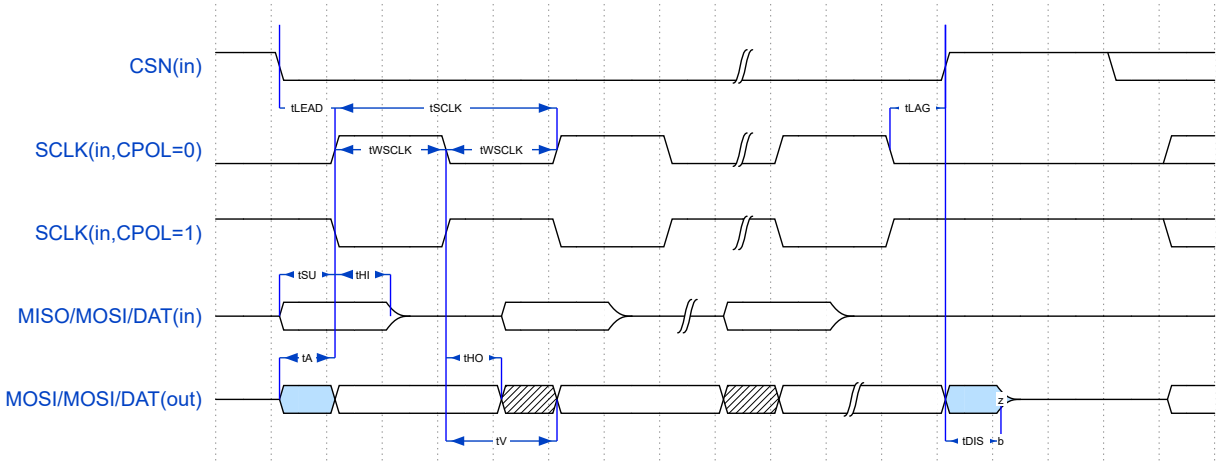



Figure 18: SPI Slave Mode Timing (CPHA=0)

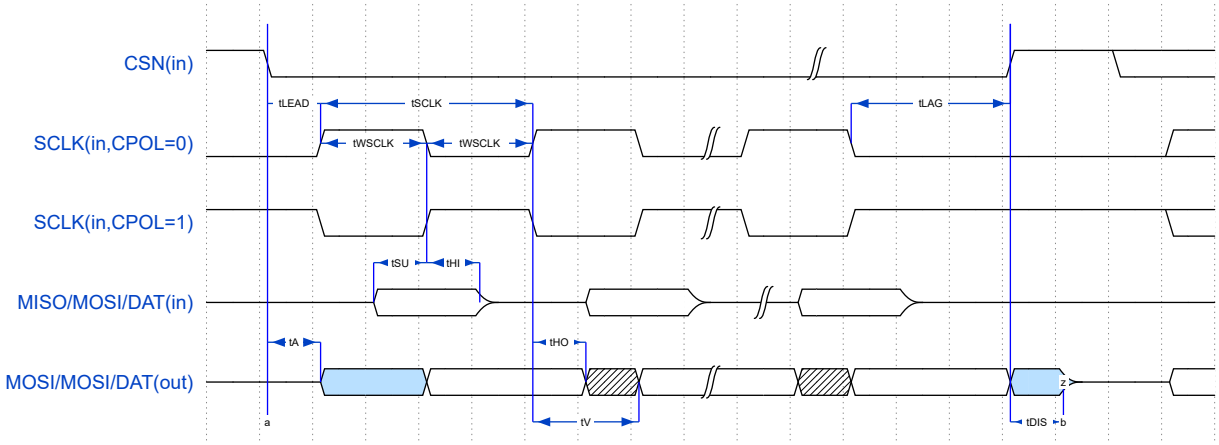



Figure 19: SPI Slave Mode Timing (CPHA=1)

| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| tSCK | SCK Period | 4 x tperiph | — | ns |
| tLEAD | CS Setup Time | 1 | — | tperiph |
| tLAG | CS Hold Time | 1 | — | tperiph |
| tWSCK | Clock High or Low Duration | tSCK / 2 - 5 | — | ns |
| tSU | Data Setup Time (Input) | 2.7 | — | ns |
| tHI | Data Hold Time (Input) | 3.8 | — | ns |

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| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| tA | Slave Access Time | — | tperiph | ns |
| tDIS | Slave MISO Disable Time | — | tperiph | ns |
| tV | Data Valid (Delay from SCLK Edge) | — | 14.5 | ns |
| tHO | Data Hold Time (Output) | 0 | — | ns |

Table 41: SPI Slave Mode Parameters (Note: tperiph = 1000 / fperiph)

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## 4.15 I2C Interface

| Symbol | Description | Operating Mode | Min. | Max. | Unit |
|---|---|---|---|---|---|
| fSCL | SCL Clock Frequency | Standard-mode (Sm) | 0 | 100 | KHz |
| | | Fast-mode (Fm) | 0 | 400 | KHz |
| | | Fast-mode Plus (Fm+) | 0 | 1000 | KHz |

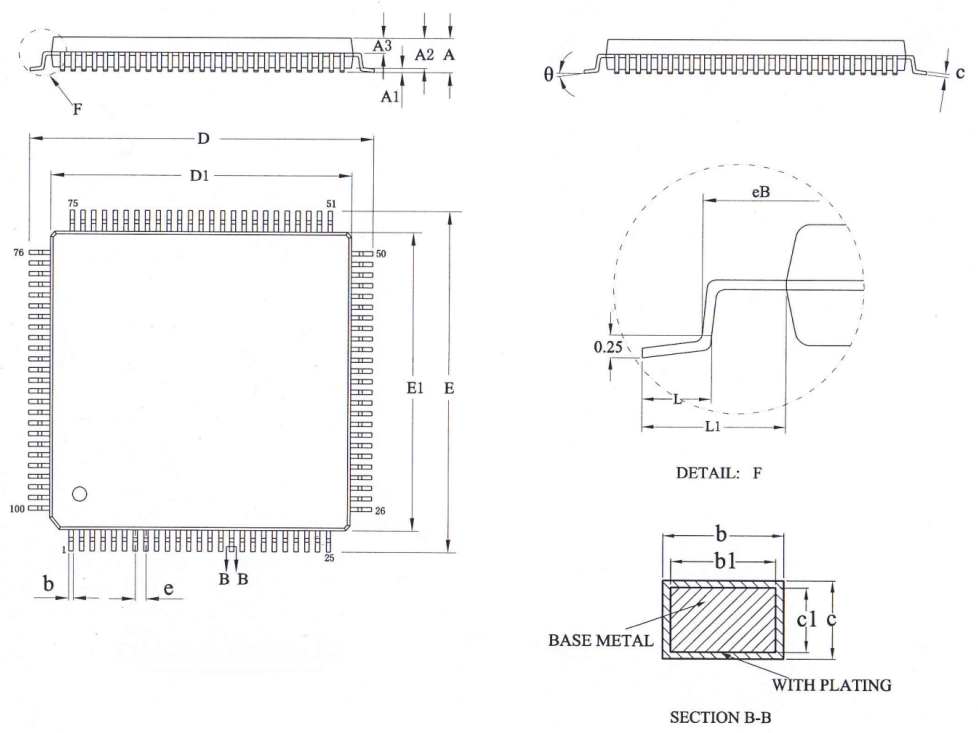
Table 42: I2C Operating Modes and Parameters

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# 5 Package

## 5.1 LQFP100 Package Dimensions

The dimensions of LQFP100 are shown in Figure 20.



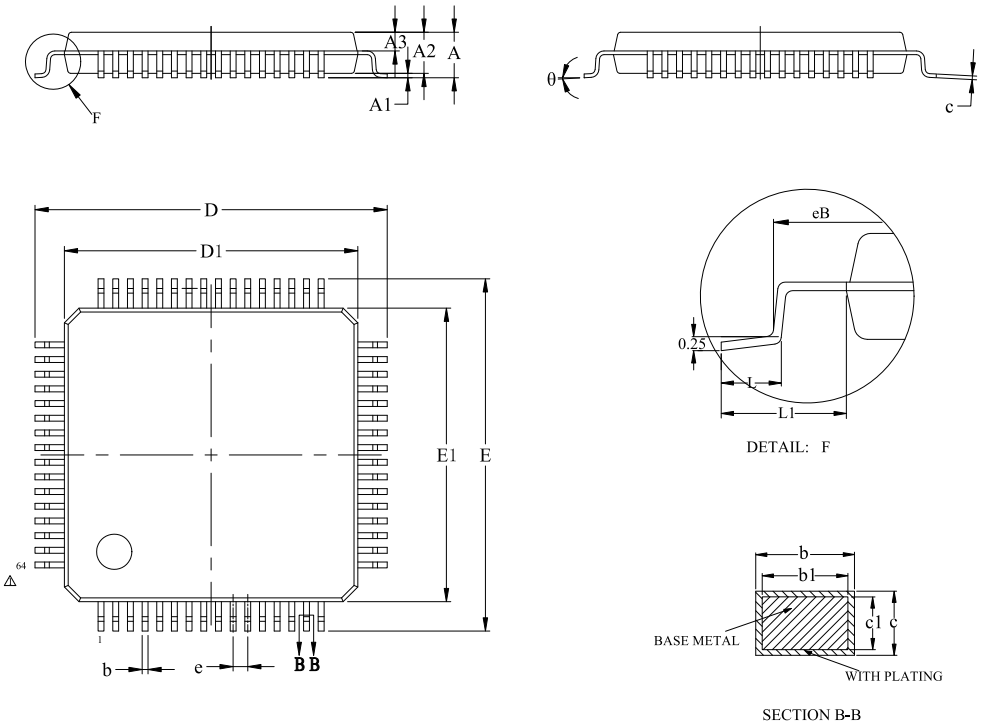

| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | — | — | 1.60 |
| A1 | 0.05 | — | 0.15 |
| A2 | 1.35 | 1.40 | 1.45 |
| A3 | 0.59 | 0.64 | 0.69 |
| b | 0.18 | — | 0.26 |
| b1 | 0.17 | 0.20 | 0.23 |
| c | 0.13 | — | 0.17 |
| c1 | 0.12 | 0.13 | 0.14 |
| D | 15.80 | 16.00 | 16.20 |
| D1 | 13.90 | 14.00 | 14.10 |
| E | 15.80 | 16.00 | 16.20 |
| E1 | 13.90 | 14.00 | 14.10 |
| eB | 15.05 | — | 15.35 |
| e | 0.50BSC | | |
| L | 0.45 | — | 0.75 |
| L1 | 1.00REF | | |
| θ | 0 | — | 7° |

Figure 20: LQFP100 Package Dimension Drawing

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## 5.2 LQFP64 Package Dimensions

The dimensions of LQFP64 are shown in Figure 21.



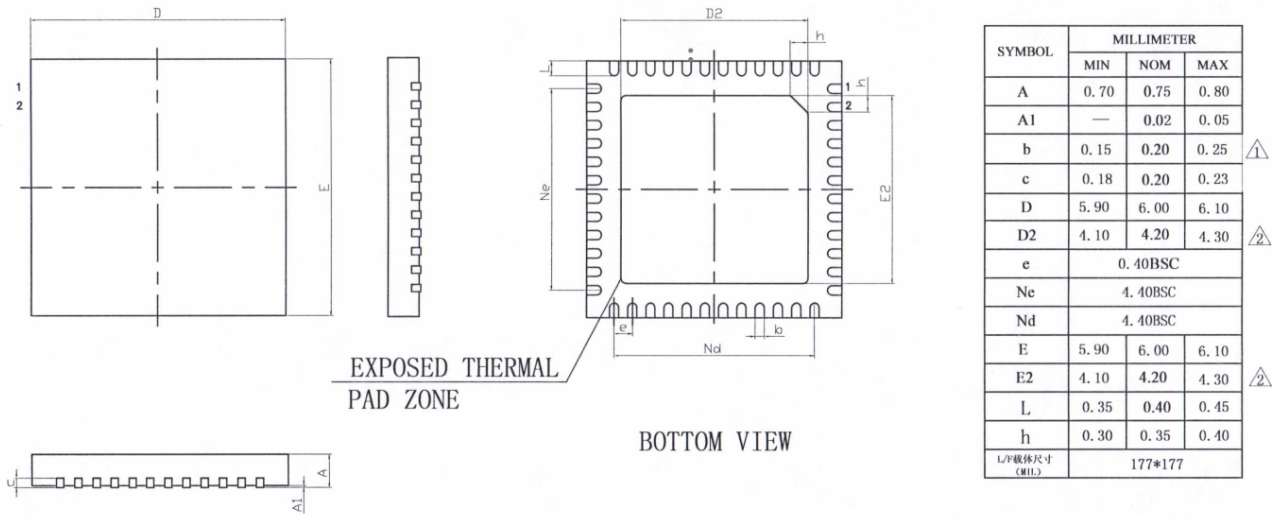

| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | — | — | 1.60 |
| A1 | 0.05 | — | 0.15 |
| A2 | 1.35 | 1.40 | 1.45 |
| A3 | 0.59 | 0.64 | 0.69 |
| b | 0.18 | — | 0.26 |
| b1 | 0.17 | 0.20 | 0.23 |
| c | 0.13 | — | 0.17 |
| c1 | 0.12 | 0.13 | 0.14 |
| D | 11.80 | 12.00 | 12.20 |
| D1 | 9.90 | 10.00 | 10.10 |
| E | 11.80 | 12.00 | 12.20 |
| E1 | 9.90 | 10.00 | 10.10 |
| e | 0.50BSC | | |
| eB | 11.05 | — | 11.25 |
| L | 0.45 | — | 0.75 |
| L1 | 1.00REF | | |
| θ | 0 | — | 7° |

Figure 21: LQFP64 Package Dimension Drawing

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## 5.3 QFN48 Package Dimensions

The dimensions of QFN48 are shown in Figure 22.




| SYMBOL | MILLIMETER | | | |
|---|---|---|---|---|
| | MIN | NOM | MAX | |
| A | 0.70 | 0.75 | 0.80 | |
| A1 | — | 0.02 | 0.05 | |
| b | 0.15 | 0.20 | 0.25 | 1 |
| c | 0.18 | 0.20 | 0.23 | |
| D | 5.90 | 6.00 | 6.10 | |
| D2 | 4.10 | 4.20 | 4.30 | 2 |
| e | 0.40BSC | | | |
| Ne | 4.40BSC | | | |
| Nd | 4.40BSC | | | |
| E | 5.90 | 6.00 | 6.10 | |
| E2 | 4.10 | 4.20 | 4.30 | 2 |
| L | 0.35 | 0.40 | 0.45 | |
| h | 0.30 | 0.35 | 0.40 | |
| L/F载体尺寸 (MIL) | 177*177 | | | |

Figure 22: QFN48 Package Dimension Drawing

## 5.4 Package Thermal Resistance Coefficient

$T_J$ max = $T_A$ max + ($P_D$ max x $\theta_{JA}$)

- $T_A$ refers to the ambient temperature during chip operation, unit: °C;
- $\theta_{JA}$ refers to the thermal resistance coefficient of the package to the operating environment, unit: °C/W;
- $P_D$ refers to the sum of the internal power consumption and I/O power consumption of the chip, unit: W;
- $T_J$ refers to the junction temperature of the chip surface.

When the chip operates at the specified ambient temperature, the internal junction temperature $T_J$ of the chip shall not exceed the maximum allowable junction temperature $T_J$ max of the chip.

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $\theta_{JA}$ | LQFP100 14X14 mm/0.5mm pitch | 50+/-5% | °C/W |
| $\theta_{JA}$ | LQFP64 10X10 mm/0.5mm pitch | 47+/-5% | °C/W |
| $\theta_{JA}$ | QFN48 6X6 mm/0.4mm pitch | 29+/-5% | °C/W |

Table 43: Thermal Resistance Coefficient Table for Each Package

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# 6 Ordering Information

## 6.1 Product Naming Convention

The product naming convention is shown in Figure 23

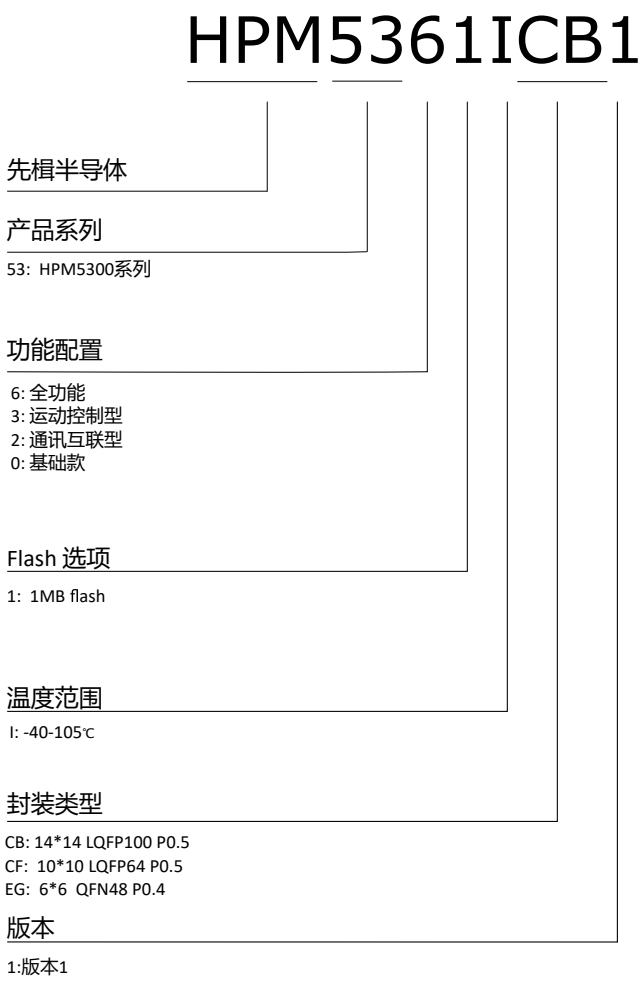



Figure 23: Product Naming Convention

## 6.2 Ordering Information

The ordering information is shown in Table 44:

| Part Number | HPM5361 | HPM5331 | HPM5321 | HPM5301 |
|---|---|---|---|---|
| CPU | RV32-IMAFDCBP | | | |
| Main Frequency (MHz) | 480 | 480 | 480 | 360 |
| Total On-chip Memory | 288 KB | | | |

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| Part Number | HPM5361 | HPM5331 | HPM5321 | HPM5301 |
|---|---|---|---|---|
| On-chip Flash | 1 MB | | | |
| ROM | 128 KB | | | |
| OTP | 4096 bit | | | |
| SDP | AES-128/256,SHA-256 | | | / |
| EXIP | XPI0: EXIP AES-128 CTR | | | / |
| RNG | True Random Number Generator | | | / |
| Secure Boot | Encrypted Boot, Trusted Boot | | | / |
| UID | 128 bit | | | |
| USB | 1 unit, integrated HS PHY | | | |
| CAN-FD | 4 | / | 4 | / |
| UART | 9 | 5(1) | 9 | 5(1) |
| SPI | 4 | 4 | 4 | 4 |
| I2C | 4 | 4 | 4 | 4 |
| PWM | 2x 8CH | | / | / |
| QEIV2 | 2 | 2 | / | / |
| QEO | 2 | 2 | / | / |
| SEI | 2 | 2 | / | / |
| MMC | 2 | 2 | / | / |
| RDC | 1 | 1 | / | / |
| PLB | 1 | 1 | 1 | / |
| TMR | 5 | | | 3(2) |
| WDG | 3 | | | |
| DMA | HDMA 32CH | | | |
| ADC | 2× 16b | | | 1× 16b(3) |
| DAC | 2× 12b | | / | / |
| CMP | 2 | | | |
| OPAMP | 2 | 2 | / | / |
| GPIO | 56 (HPM53XXICB1), 36 (HPM53XXICF1), 29 (HPM53XXIEG1) | | | |
| Package | 14*14 LQFP100 P0.5 (HPM53XXICB1)<br>10*10 LQFP64 P0.5 (HPM53XXICF1)<br>6*6 QFN48 P0.4 (HPM53XXIEG1) | | | |
| Temperature Range $T_A$ | -40~105 °C | | | |

Table 44: Ordering Information

(1) UART0, UART1, UART2, UART3 and PUART are available.

(2) GPTMR0, GPTMR1 and PTMR are available.

(3) ADC0 is available.

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## 6.3 Functional Differences Between Packages

The functional differences between different packages of this product are shown in Table 45.

| | HPM53xxxCBx | HPM53xxxCFx | HPM53xxIEGx |
|---|---|---|---|
| Package | LQFP100<br>14mm×14mm<br>p0.5mm | LQFP64<br>10mm×10mm<br>p0.5mm | QFN48<br>6mm×6mm<br>p0.4mm |
| GPIO | 56 | 36 | 29 |
| Analog Function Pins | 16 | 10 | 8 |
| Communication Interfaces | 9x UART, 4x SPI, 4x I2C | 7x UART, 4x SPI, 4x I2C<br>UART4, UART5 not brought out | 7x UART, 3x SPI, 4x I2C<br>UART4, UART5, SPI3 not brought out |
| BOOT ROM ISP | USB/UART0 | UART0 | UART0 |
| Internal IRC24M Boot | Not Supported | Supported(1) | Supported(1) |

Table 45: Functional Differences Between Packages

(1) The MCU is booted from the internal oscillator IRC24M, which is used as the reference clock input for the PLL. If required, the user shall manually enable the external crystal OSC24M and switch the PLL reference clock.

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# 7 Revision Information

| Date | Revision | Description |
|---|---|---|
| Rev0.0 | 2023/05/12 | Rev0.0 released. |
| Rev0.1 | 2023/05/19 | Rev0.1 released.<br>Adjusted product model information.<br>Updated content of the introduction section.<br>Updated system block diagram. |
| Rev0.2 | 2023/06/25 | Rev0.2 released.<br>Adjusted product feature information on the cover.<br>Updated content of the introduction section.<br>Updated IO reset status table.<br>Updated IDD current characteristic table.<br>Updated product selection table. |
| Rev0.3 | 2023/07/28 | Rev0.3 released.<br>Updated content of the introduction section.<br>Added OPAMP characteristic table.<br>Updated ballmap diagram.<br>Updated built-in flash features. |
| Rev0.4 | 2023/08/15 | Rev0.4 released.<br>Updated product model information.<br>Updated product ordering information table.<br>Updated power consumption current data. |
| Rev0.5 | 2023/08/31 | Rev0.5 released.<br>Updated product ordering information table.<br>Updated ADC electrical characteristics. |
| Rev0.6 | 2023/10/27 | Rev0.6 released.<br>Updated product ordering information table.<br>Updated product model information on the cover.<br>Updated IO reset status, added description of IO power supply pins.<br>Updated partial descriptions in the power supply section. |
| Rev0.7 | 2024/03/08 | Rev0.7 released.<br>Corrected description of the motion control system part in the product introduction section.<br>Added description of reset pin characteristics.<br>Corrected description of the IO AC characteristic table. |
| Rev0.8 | 2024/04/17 | Rev0.8 released.<br>Corrected partial descriptions in the ordering information section. |

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| Date | Revision | Description |
|---|---|---|
| Rev0.9 | 2024/05/15 | Rev0.9 released.<br>Added peripheral clock information. |
| Rev0.10 | 2024/08/01 | Rev0.10 released.<br>Corrected the error in the function list of PINMUX PA12 and PA13. |
| Rev0.11 | 2025/01/15 | Rev0.11 released.<br>Refined the correspondence between frequency and voltage under different working scenarios.<br>Corrected the pin pitch of the QFN48 package. |
| Rev0.12 | 2025/11/20 | Rev0.12 released.<br>Added power supply block diagram under DCDC BYPASS scenario. |

Table 46: Revision Information

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# 8 Disclaimer

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